

Self-Supervised Learning based Speech Representations

DEEP SPEECH PROCESSING Course

IIIT Dharwad

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Module: Self-Supervised Learning based Speech Representations

1. **Representation learning for speech:** a brief historical context.
2. **Self-Supervised Learning (SSL):** Motivation: reducing labeled data requirements, leveraging large unlabeled corpora. General frameworks: contrastive, generative, and predictive.
3. **SSL-based Pretraining:** Mockingjay, wav2vec 2.0, HuBERT, etc.
4. **Fine-tuning and domain adaptation:** Fine-tuning SSL models for downstream tasks, resource-constrained adaptation (low-resource languages, few-shot learning)
5. **Benchmarking:** To understand information encoded in speech representations and to, evaluate and compare their performance. For example, SUPERB, IndicSUPERB, etc.

Key references:

1. *Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, [arXiv:2110.09327](https://arxiv.org/abs/2110.09327)*
2. *Abdelrahman Mohamed, et al., Self-Supervised Speech Representation Learning: A Review, arxiv.2205.10643*
3. *A. Baevski, et al., "wav2vec 2.0: A framework for self-supervised learning of speech representations," Proceedings of NeurIPS, vol. 33, 2020.*
4. *W.-N. Hsu, et al., "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," arXiv preprint arXiv:2106.07447, 2021.*
5. *Javed, Tahir, et al. "Indicsuperb: A speech processing universal performance benchmark for Indian languages." Proceedings of AAAI Conference on Artificial Intelligence. Vol. 37. No. 11. 2023.*
6. *Arora, Siddhant, et al. "On the Evaluation of Speech Foundation Models for Spoken Language Understanding." arXiv preprint arXiv:2406.10083 (2024).*

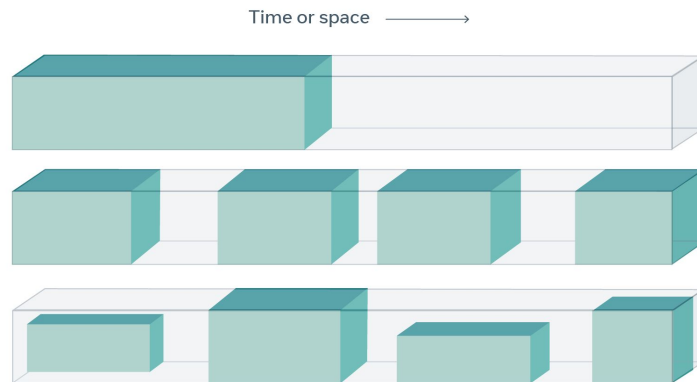
What is self-supervised learning?

Self-supervised learning obtains supervisory signals from the data itself

Labels are naturally part of the input data

Predict any part of the input from any other part

Build foundation model to learn robust representations from vast amounts of unlabeled data



[1] Image source: [Yann LeCun, Ishan Misra, "Self-supervised learning: The dark matter of intelligence", Meta AI](#)

[2] [A Cookbook of Self-Supervised Learning](#)

Foundation models

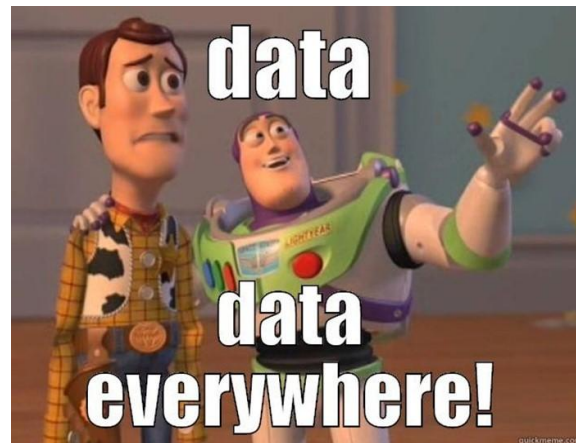
- Developing well-performing DL models has become *costly, involving data collection, modeling, computing power, and training time*.
- Need to repeat for each specific use case: time-consuming and cost-prohibitive.
- **Foundation models** are DL models trained on huge amounts of **unlabeled datasets** (audio-only data) that can be used to solve multiple downstream tasks with minimal labeled data.
- This approach has proven crucial in the field of Natural Language Processing (NLP) - **GPT, BERT**

SSL Motivation

DL techniques depends on the quality of the representations automatically discovered from data

These representations must capture important underlying structures from the raw input, e.g., intermediate concepts, features - useful for the downstream task.

Supervised representations might be biased towards a specific application.



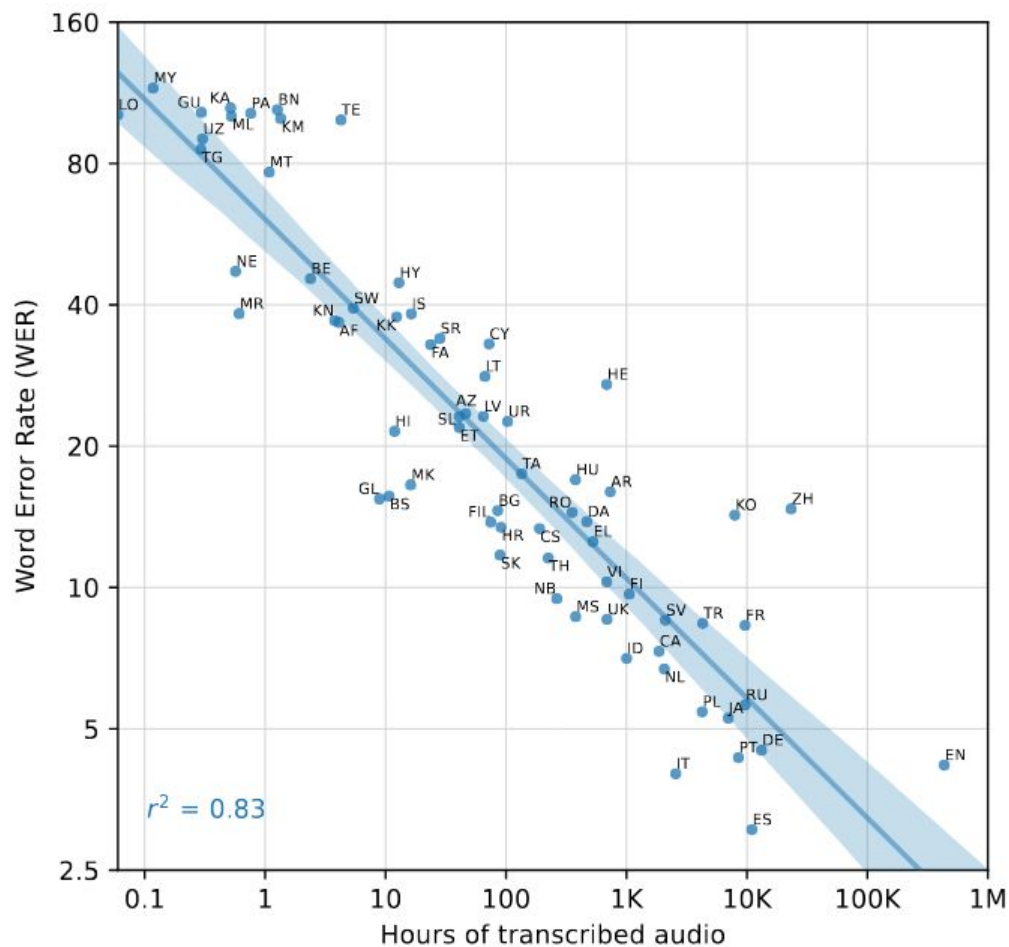
Collecting large amounts of annotated examples is costly

Unavailability of data for low-resource applications

1. Yoshua Bengio. [Deep learning of representations for unsupervised and transfer learning](https://icml-sas.gitlab.io/#fn:bengio2012deep), 2012
2. <https://icml-sas.gitlab.io/#fn:bengio2012deep>

Motivation:

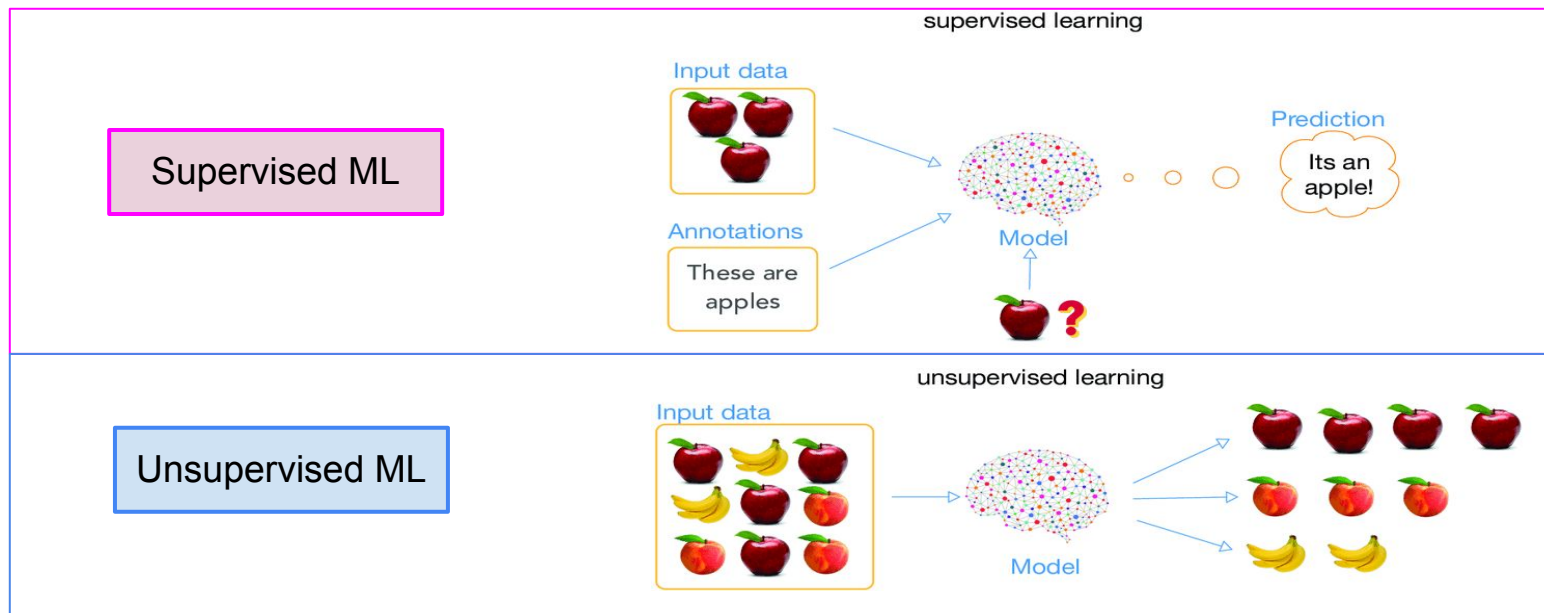
Radford, Alec, et al. "Robust speech recognition via large-scale weak supervision." *International conference on machine learning*. PMLR, 2023.



What is machine learning?

Machine learning (ML) algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to do so^[1,2].

- Let's consider **two of categories of ML**:



1. https://en.wikipedia.org/wiki/Machine_learning
2. [Arthur Samuel](#), who coined the term "machine learning" in 1959
3. Image source: DOI:[10.3390/sym10120734](https://doi.org/10.3390/sym10120734)

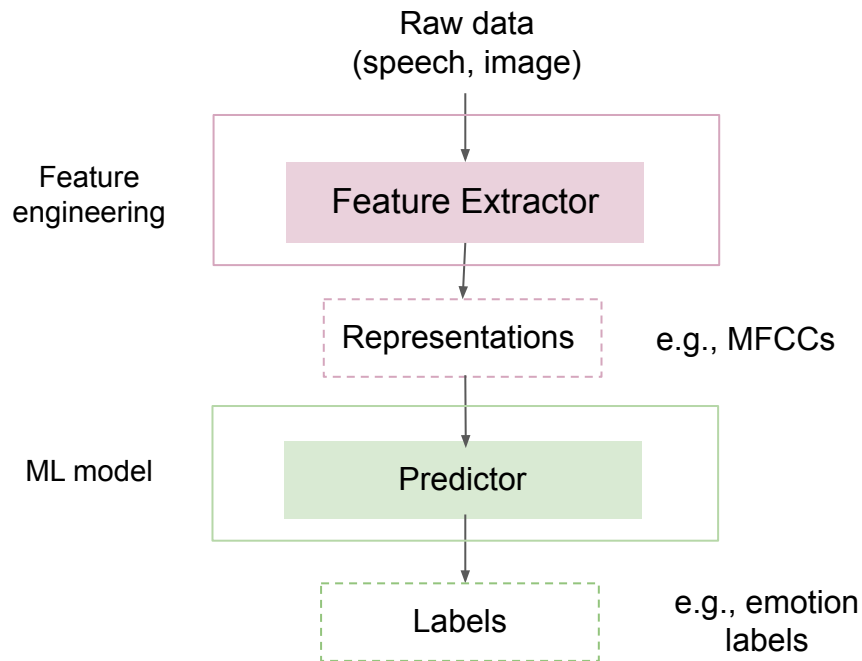
Supervised learning

Input	Output	Application
Email	Spam (Yes/No)	Spam Classifier
Speech	Text	Speech recognition
Hindi text	Assamese Text	Machine Translation
House in Sq Ft	House price (continuous value)	House price prediction
Customer review of a product	Sentiment class (Positive/Negative)	Customer's sentiment analysis of a product
Atmospheric variables like cloud cover, humidity, wind, and average temperature	Rainfall (in mm) - (continuous value)	Rainfall prediction (in mm)
Soil, water and air quality	Plant growth (in cm)	Effects of air or water pollution on plant health.

Issues with supervised learning

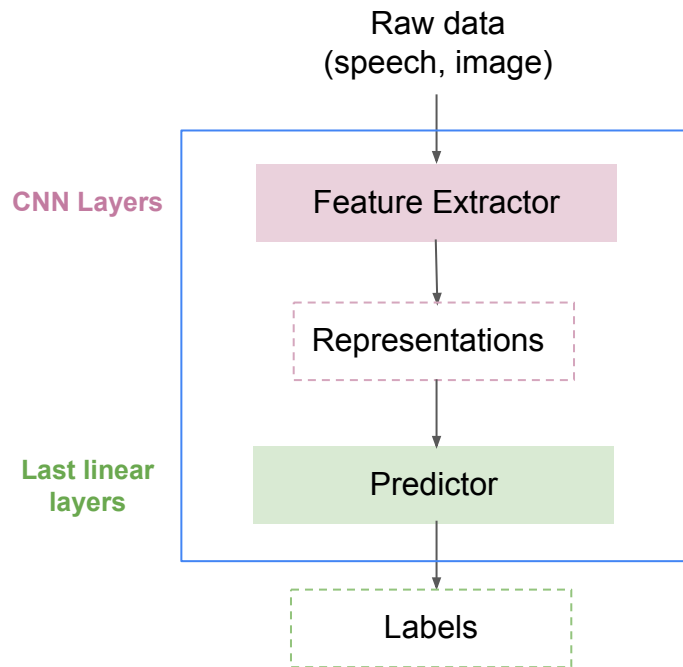
- Supervised learning relies heavily on large volumes of high-quality labeled data - huge limitation in the domains like medical imaging.
- Class imbalance adversely affects the model performance.
- Acquiring enough data for rare classes is difficult—for example, data for a newly identified wild species of birds.

Pipelines of doing ML



Typical flow

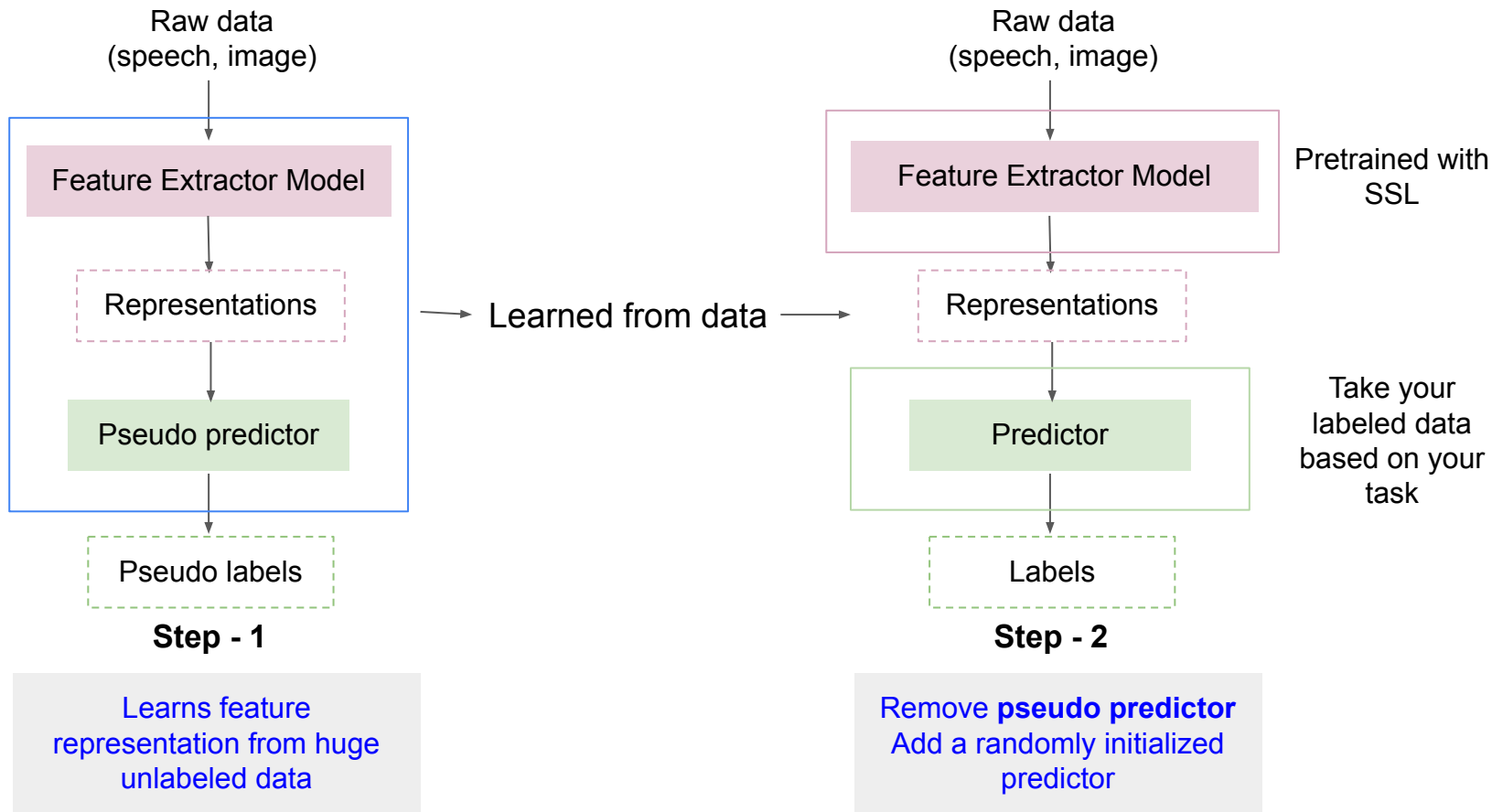
Learns from <data, label> pairs
Handcrafted features



End-to-end

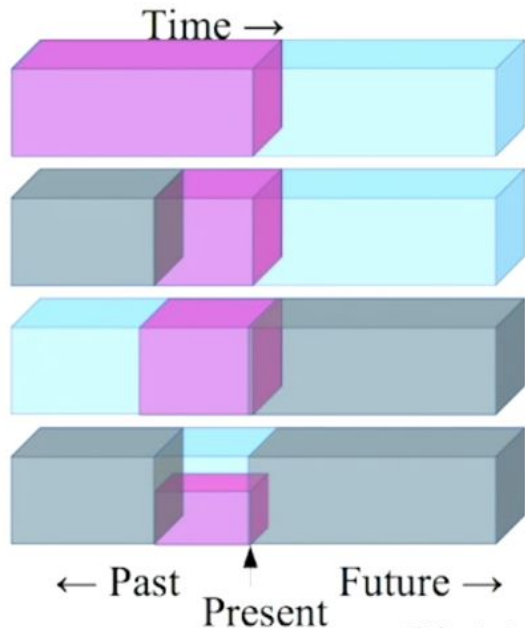
Learns from <data, label> pairs
Jointly learn feature extractor

Self-supervised Learning (SSL) - based pretraining - idea



Generating pseudo labels

- ▶ Predict any part of the input from any other part.
- ▶ Predict the **future** from the **past**.
- ▶ Predict the **future** from the **recent past**.
- ▶ Predict the **past** from the **present**.
- ▶ Predict the **top** from the **bottom**.
- ▶ Predict the occluded from the visible
- ▶ **Pretend there is a part of the input you don't know and predict that.**



Slide: LeCun

Two tasks

Pretext task or auxiliary task or pre-training task:

- > Learning through pretext task optimization
- > Train the model on huge unlabeled data to learn general representations.

Fine-tuning task or downstream task:

- > Use the representations (embeddings) learned by performing the pre-training as a starting point for the downstream supervised tasks.

Downstream task: the task that you actually want to solve.

SSL formulation for NLP

- Center Word Prediction - Word2vec - learning representations

A quick brown fox jumps over the lazy dog

- **Auto-regressive Language Modeling - GPT paper - generating text**

Nothing is _____

- **Masked Language Modeling - Google BERT - learning representations**

Randomly masked: A quick [MASK] fox jumps over the [MASK] dog

Predict: A quick brown fox jumps over the lazy dog

- **Next Sentence Prediction - Google BERT - learning representations**

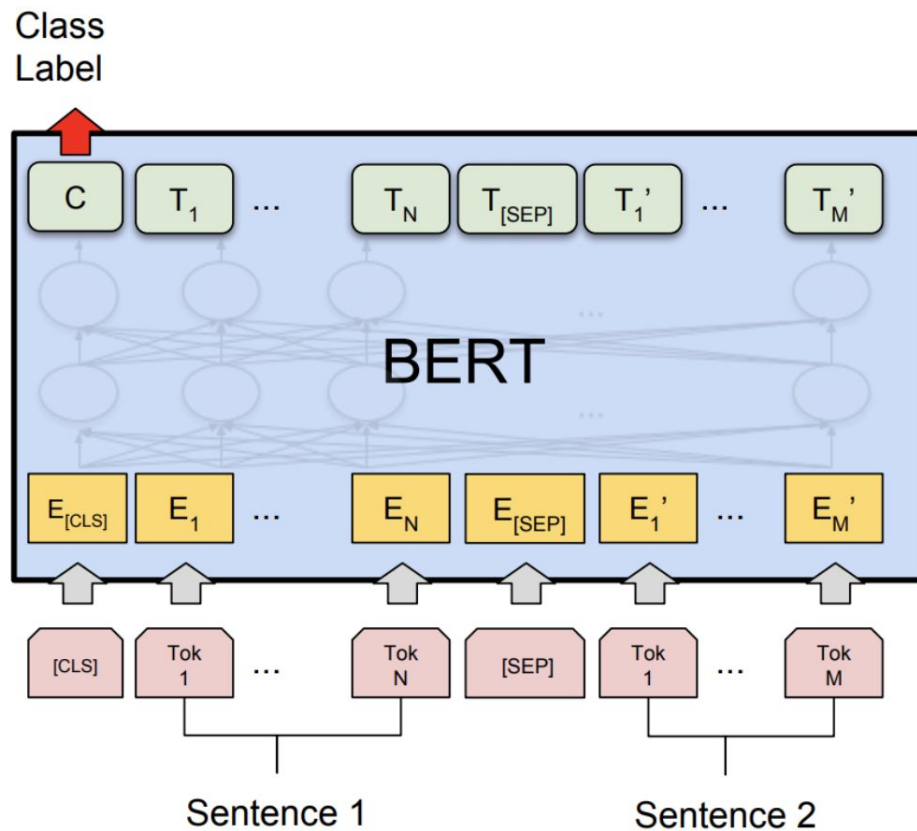
Sentence 1	Sentence 2	Next Sentence
I am going outside	I will be back in the evening	yes
I am going outside	You know nothing John Snow	no

Generating training data for Auto-regressive LM

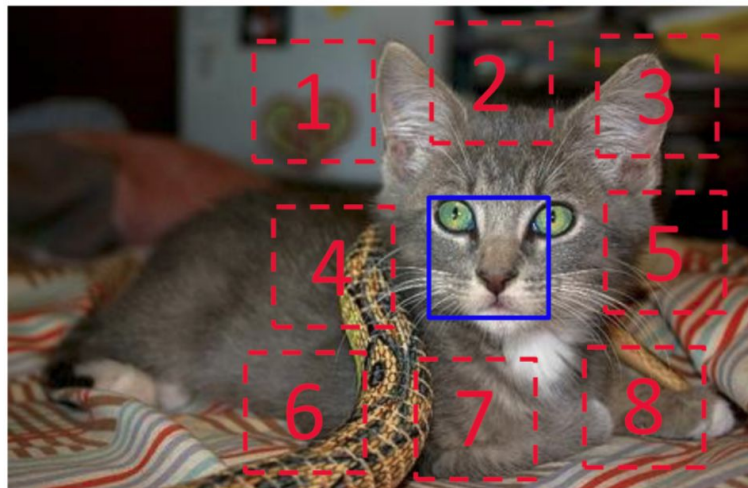
i am studying speech technology.

input	output
<bos>	i
i	am
i am	studying
i am studying	speech
i am studying speech	technology
i am studying speech technology	.
i am studying speech technology.	<eos>

BERT training

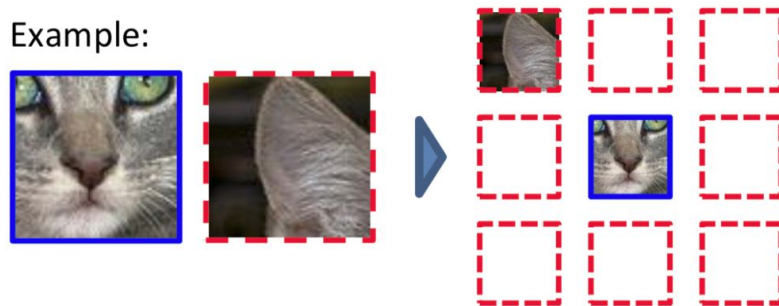


Vision: Placing image patches in right place



$$X = \left(\begin{array}{c} \text{[Kitten Face Patch]} \\ \text{[Kitten Ear Patch]} \end{array} \right); Y = 3$$

Example:



Question 1:

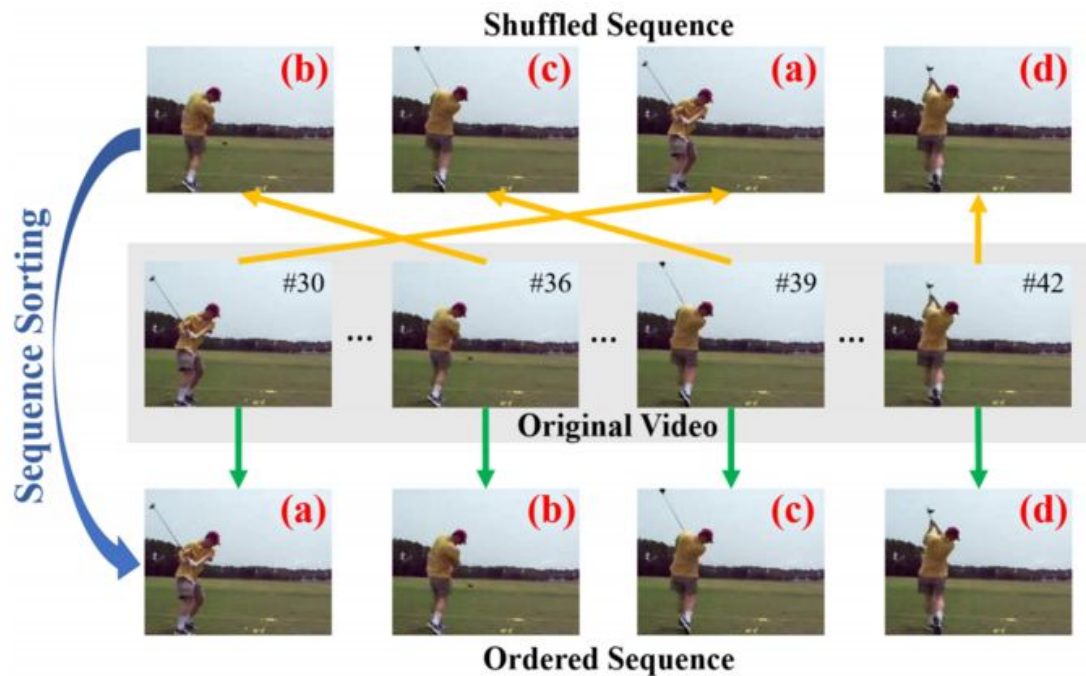


Question 2:



Figure: Illustration of self-supervised learning by predicting the relative position of two random patches. (Image source: [Doersch et al., 2015](#). Answer key: Q1: Bottom right, Q2: Top center)

Vision: placing frames in the right sequence

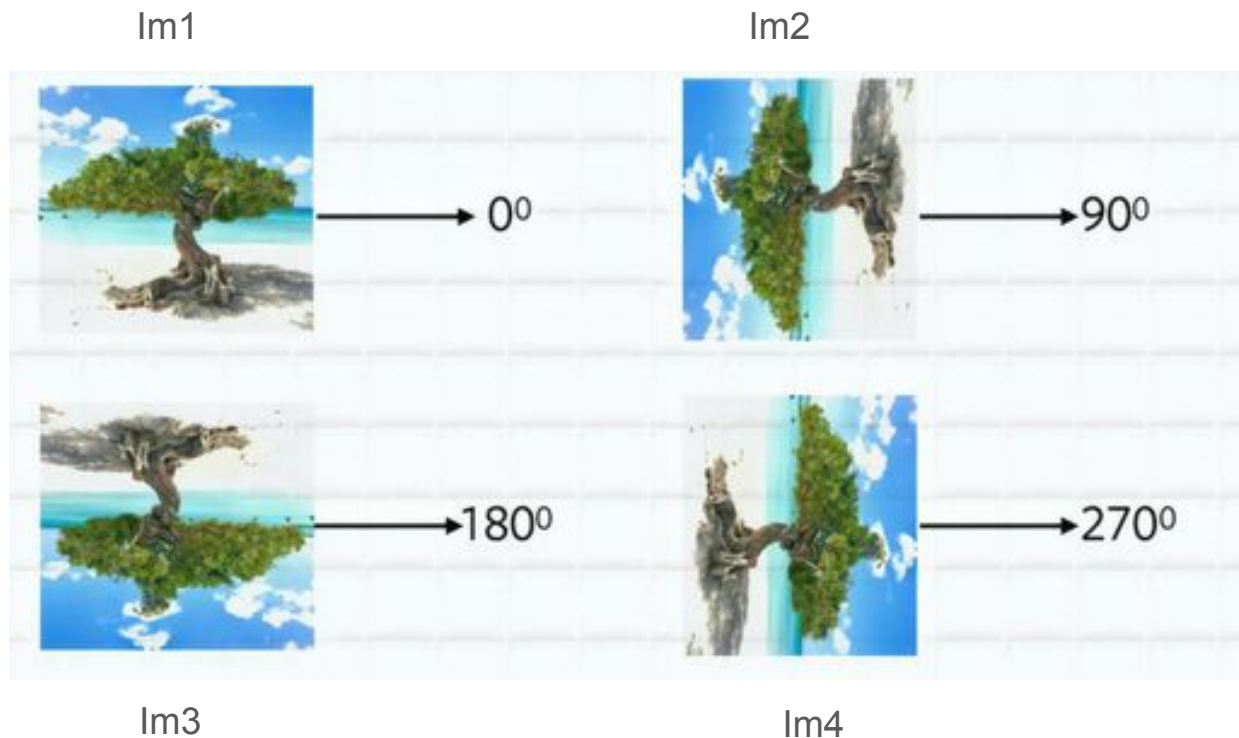


1. Hsin-Ying Lee, et. al., "Unsupervised Representation Learning by Sorting Sequences", arXiv, 2017
2. Ishan Misra, et. al., "Shuffle and Learn: Unsupervised Learning using Temporal Order Verification", arXiv, 2016

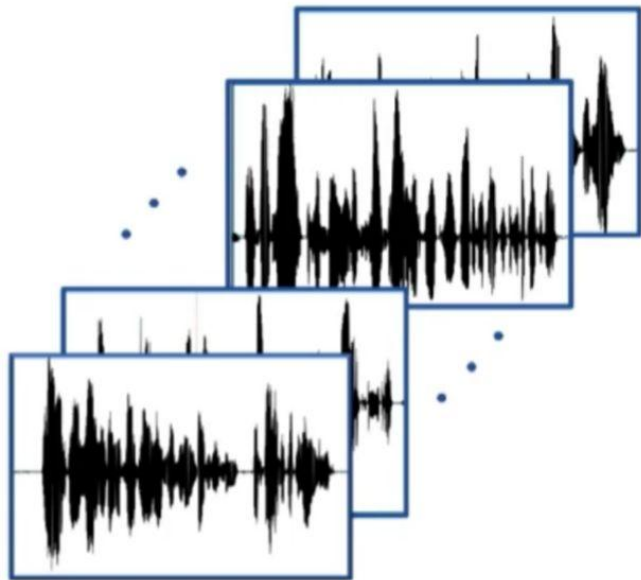
Vision: predicting rotation of image as a pretext task

Training data

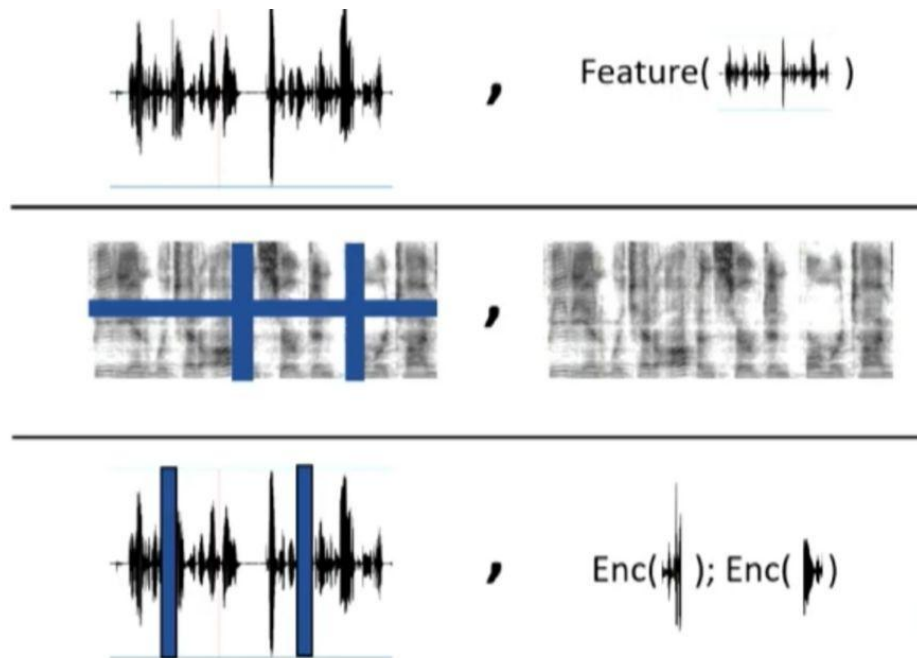
Input	Output
Im1	0
Im2	90
Im3	180
Im4	270



SSL formulation in speech

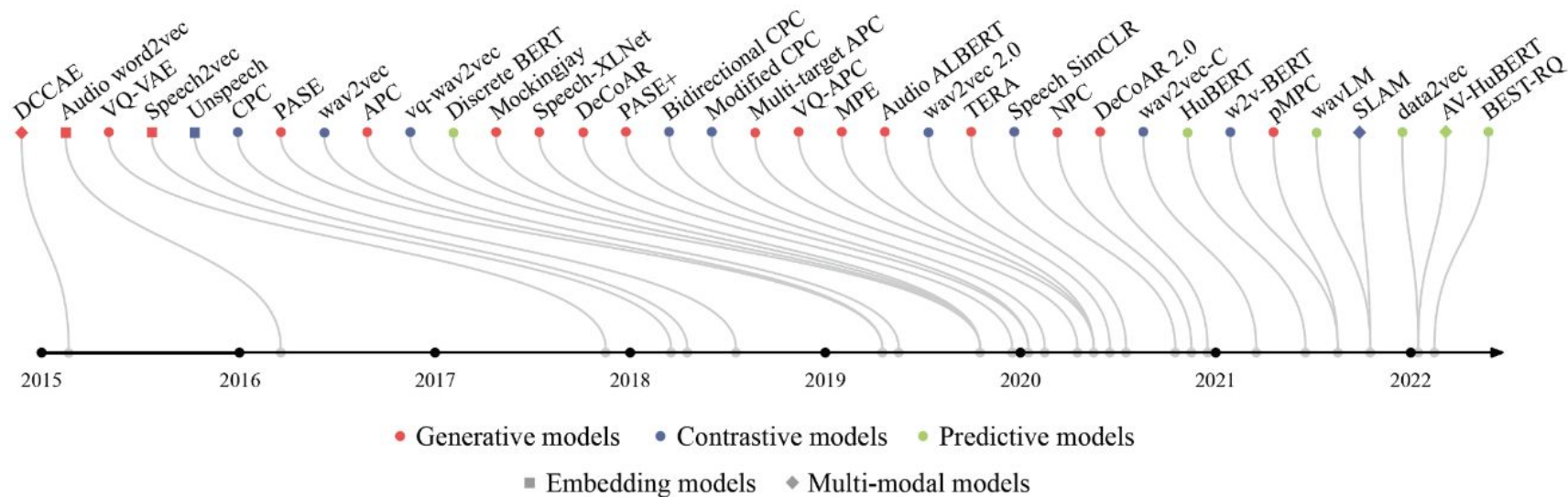


Input: Unlabeled (raw) speech audio



Self-supervised pretext tasks

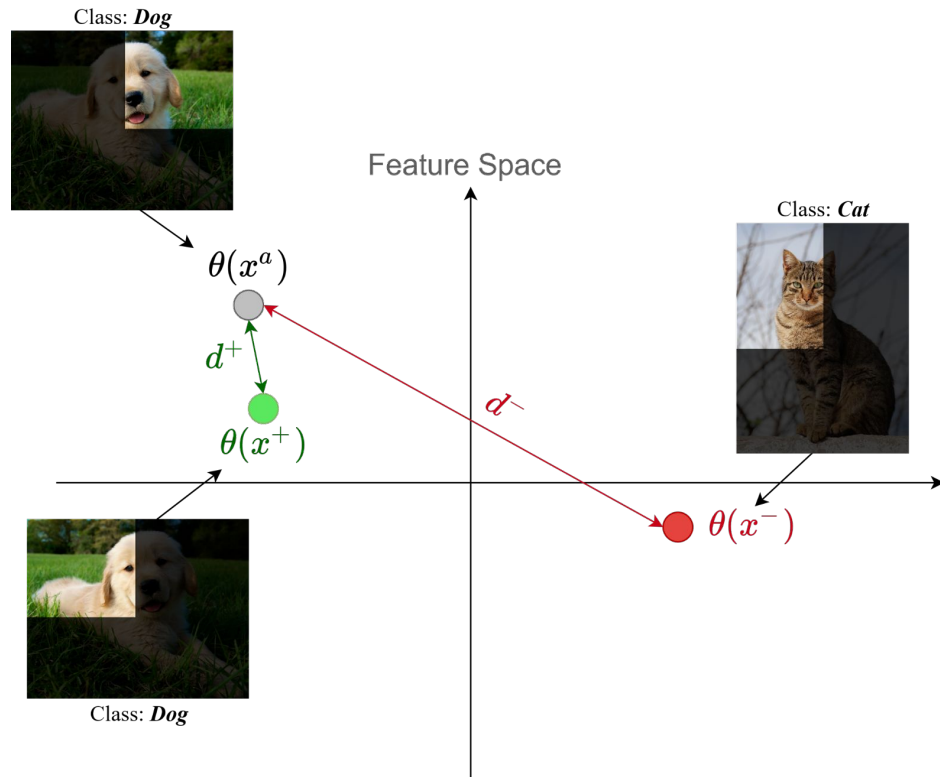
SSL models for speech



[Self-Supervised Speech Representation Learning: A Review](#)

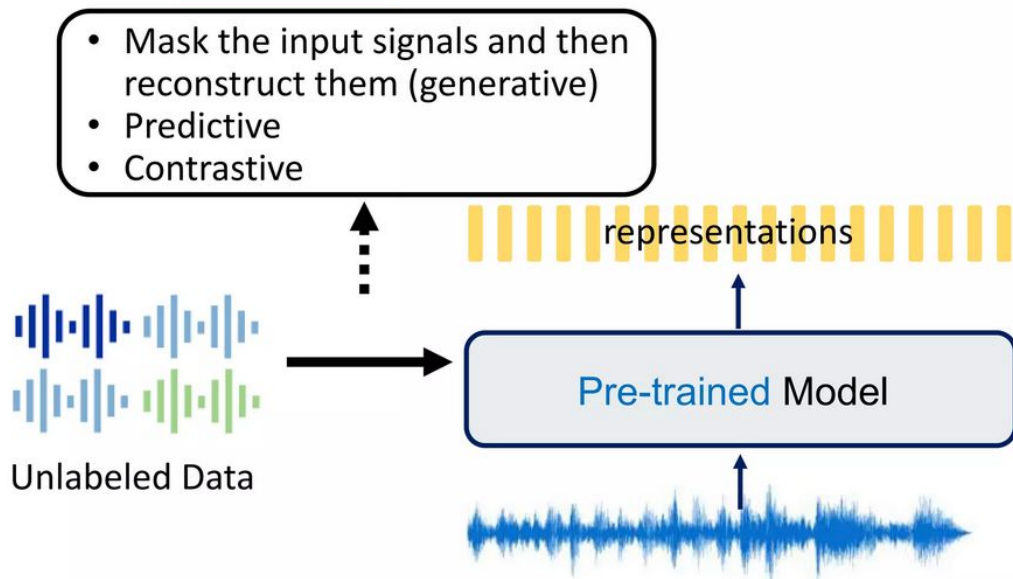
Contrastive loss

- Teach a model to pull together the representations of a target image (a.k.a., the “**anchor**”) and a matching (“**positive**”) image in embedding space, while also pushing apart the anchor from many non-matching (“**negative**”) images.



SSL for speech

Phase 1: Pre-training



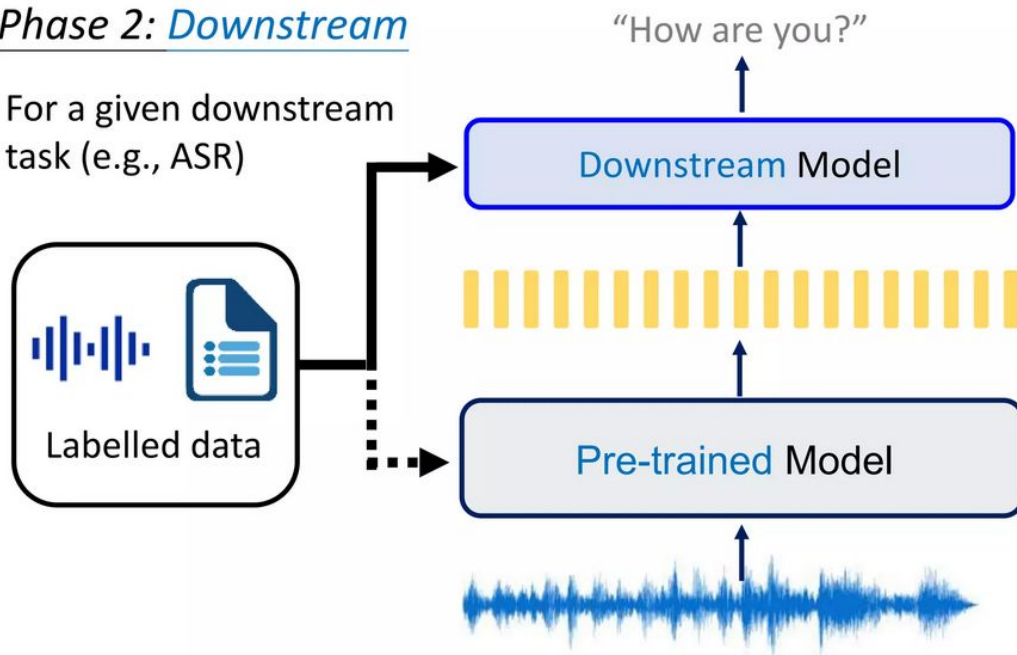
- To learn the **hidden structure** within speech signals without considering any specific downstream task

[IEEE JSTSP 2022]

SSL for speech

Phase 2: Downstream

For a given downstream task (e.g., ASR)



- With the **hidden structure** learned from the signals, the given task becomes easier

[IEEE JSTSP 2022]

We will discuss

Wav2Vec 2.0: Meta AI

CPC

HuBERT: Google

WavLM: Microsoft

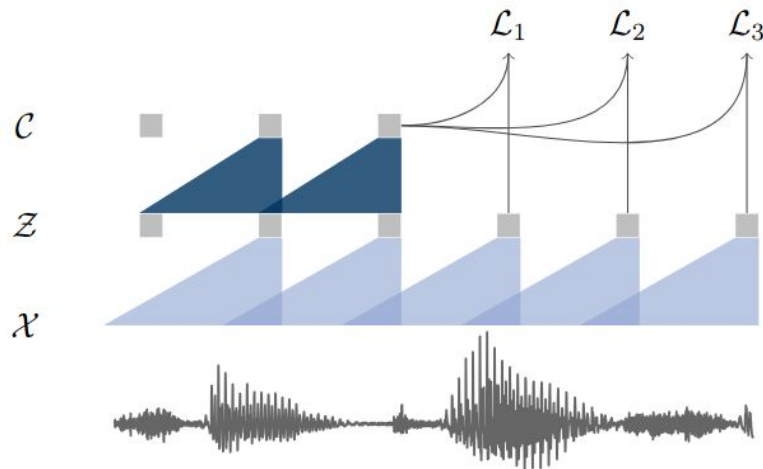
WAV2VEC: UNSUPERVISED PRE-TRAINING FOR SPEECH RECOGNITION

CNN Encoder: 1D CNN - learning **continuous latent representation** z (encoding about 30ms with 10ms step).

Context Network: Another CNN - taking sequences of the latent features, capturing some temporal context (encoding about 210ms).

Contrastive loss that requires distinguishing a true future audio sample from negatives

Analogous to LM, but instead of predicting the next word in a sentence, wav2vec tries to predict the next hidden feature vector.



Steffen Schneider, Alexei Baevski, Ronan Collobert, and Michael Auli. wav2vec: Unsupervised pre-training for speech recognition. CoRR, abs/1904.05862, 2019. URL: <http://arxiv.org/abs/1904.05862>.

Wav2vec

negatives	dev PER	train time (h)
1	16.3	6.1
2	15.8	6.3
5	15.9	8.2
10	15.5	10.5
20	15.7	15.3

Table 3: Effect of different number of negative samples during pre-training for TIMIT on the development set.

wav2vec

VQ-WAV2VEC: SELF-SUPERVISED LEARNING OF DISCRETE SPEECH REPRESENTATIONS

Learn continuous speech representations in a SSL way via predicting context information

Learning discrete representations of speech

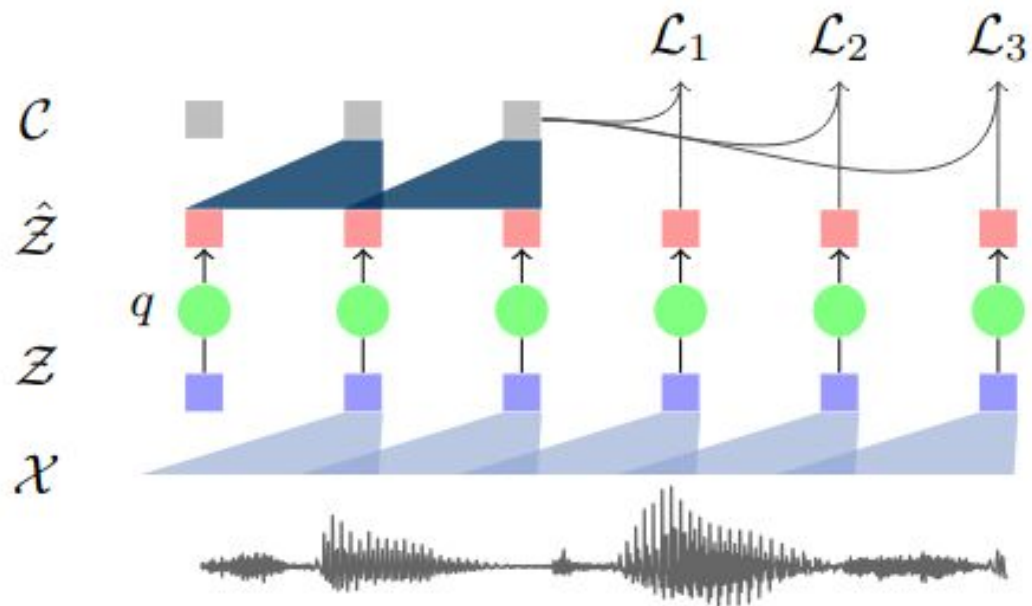
vq-wav2vec

Two CNNs same as wav2vec - for feature extraction and aggregation

Quantization: It applies **vector quantization** to the continuous latent representations.

Discrete Units: After the 1st CNN encoder, the continuous latent representations are mapped to a finite set of discrete codebook entries (similar to how discrete tokens represent words in text).

VQ-WAV2VEC



(a) vq-wav2vec

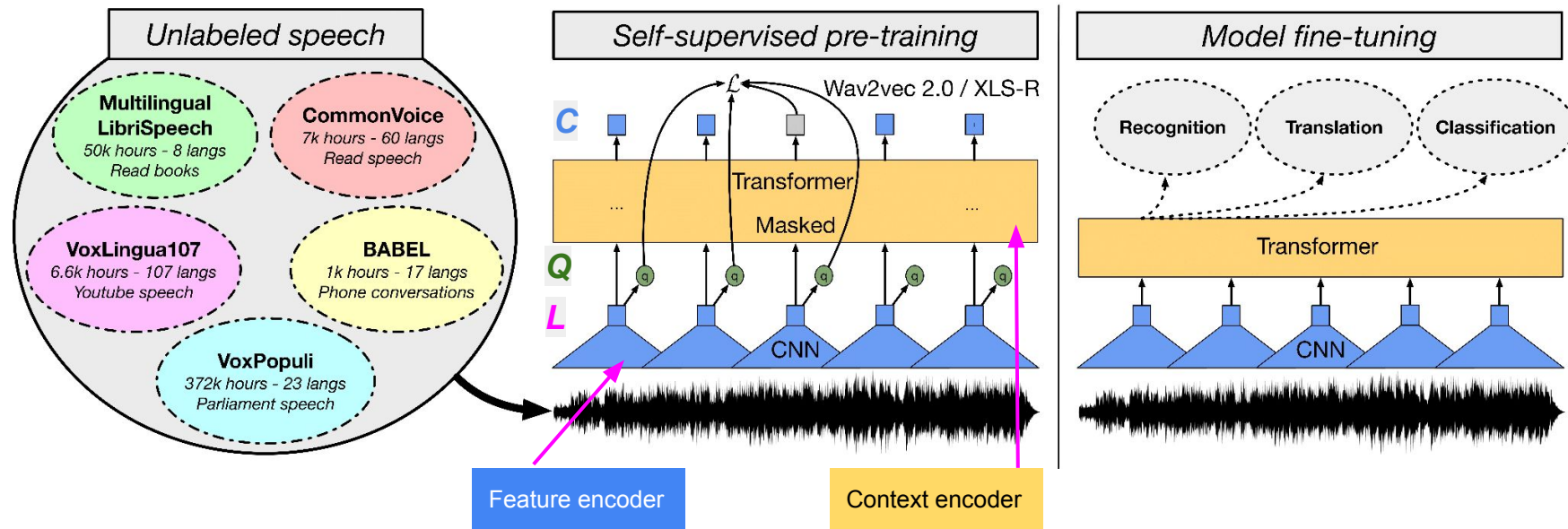
Wav2vec2

- Learns powerful speech representations from large amount of unlabeled speech
- The task: predict masked speech frames
- Facebook pretrained model:
 - <https://github.com/pytorch/fairseq/blob/master/examples/wav2vec2/README.md>)
- Contrastive loss: identify the true quantized latent speech representation for a masked time step
- **Cross-lingual training**

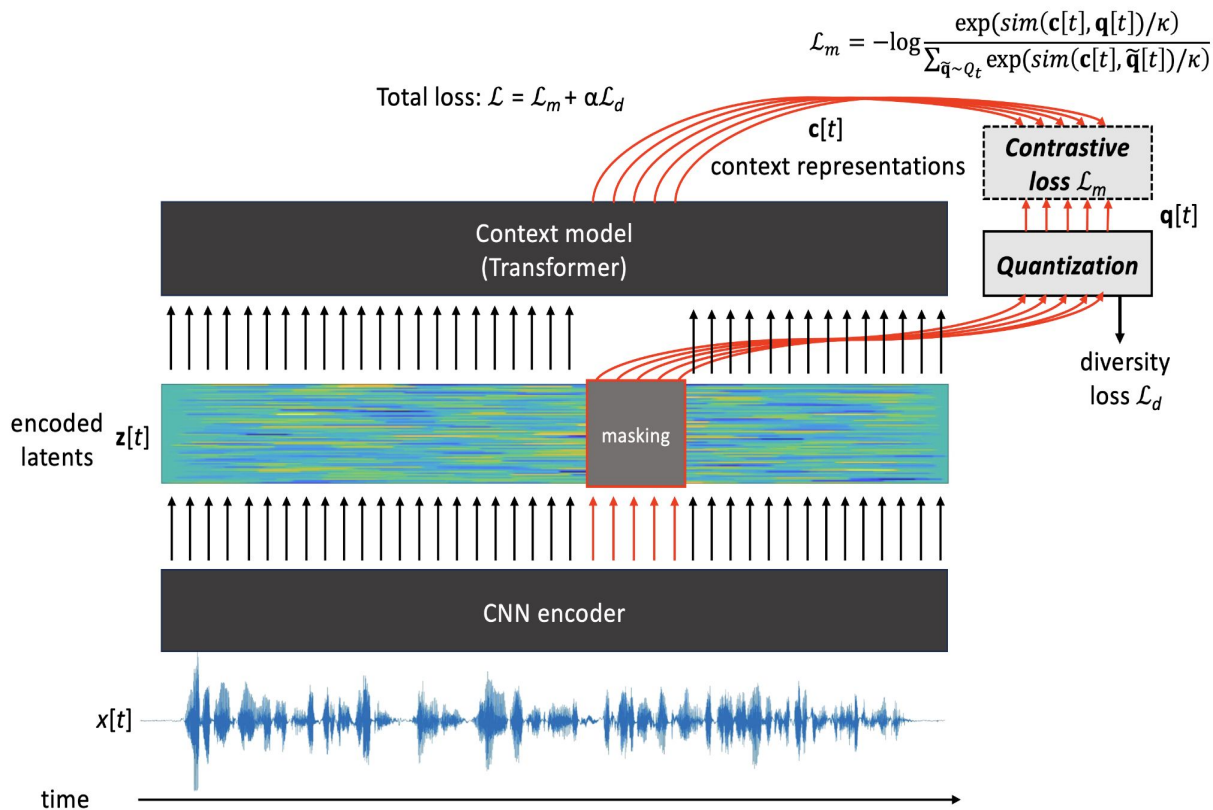
Wav2vec2 - architecture

- <https://papertalk.org/papertalks/9599> - listen before the session

Context representation: **C**
Latent representation: **L**
Quantized representation: **Q**

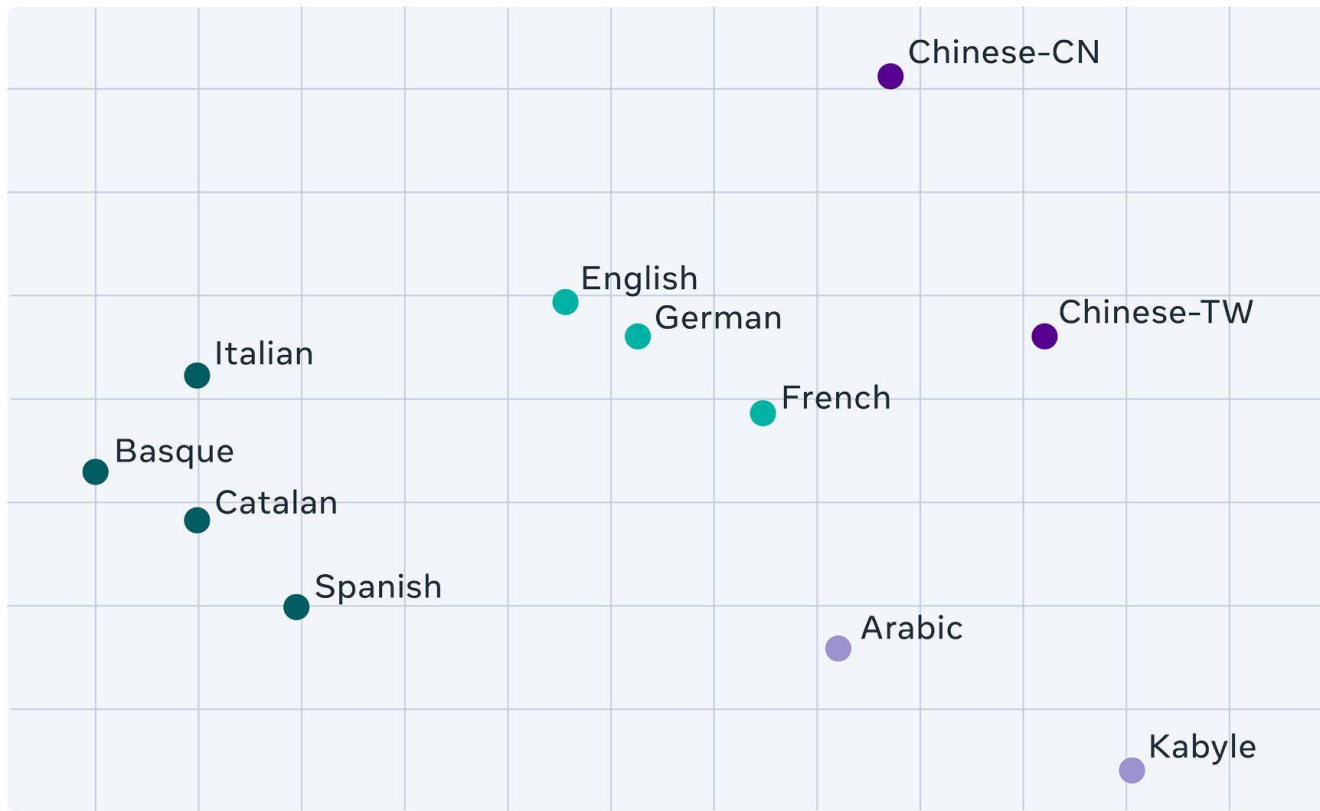


1. Alexei Baevski, et. al, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations", arXiv, 2020.
2. Image source: <https://huggingface.co/blog/fine-tune-xlsr-wav2vec2>



Masking based models attempt to predict parts of input data that are masked (hidden) from the network.

Task: is to output correct predictions $c[t]$ for the masked segments by using the surrounding context. The masked $z[t]$ are first vector quantized (VQ) into $q[t]$ with a learnable codebook, making the prediction targets categorical in nature.



Visualization of how the learned units are used across languages. Graph shows a 2D PCA plot of how units are used in each language. Languages closer to each other, like English and German or Basque and Catalan, tend to use similar units.

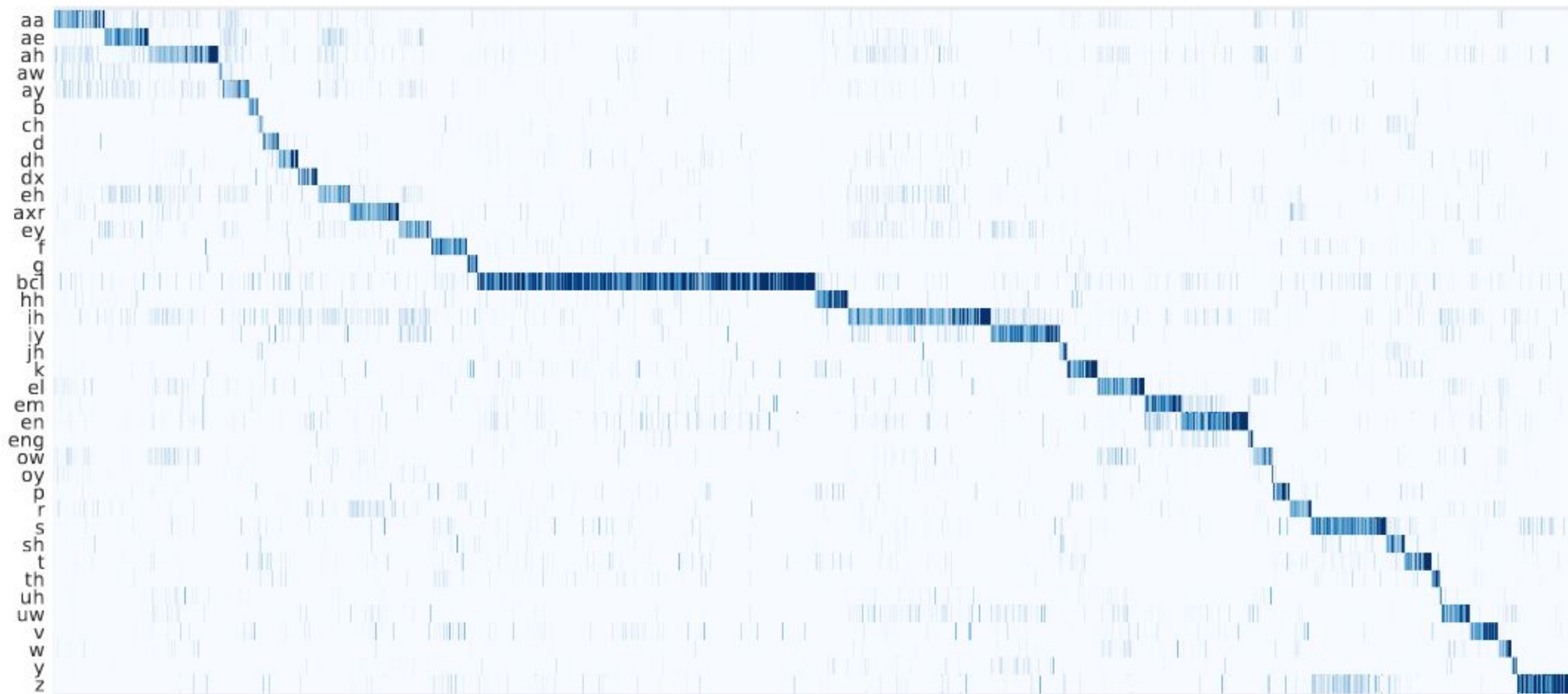
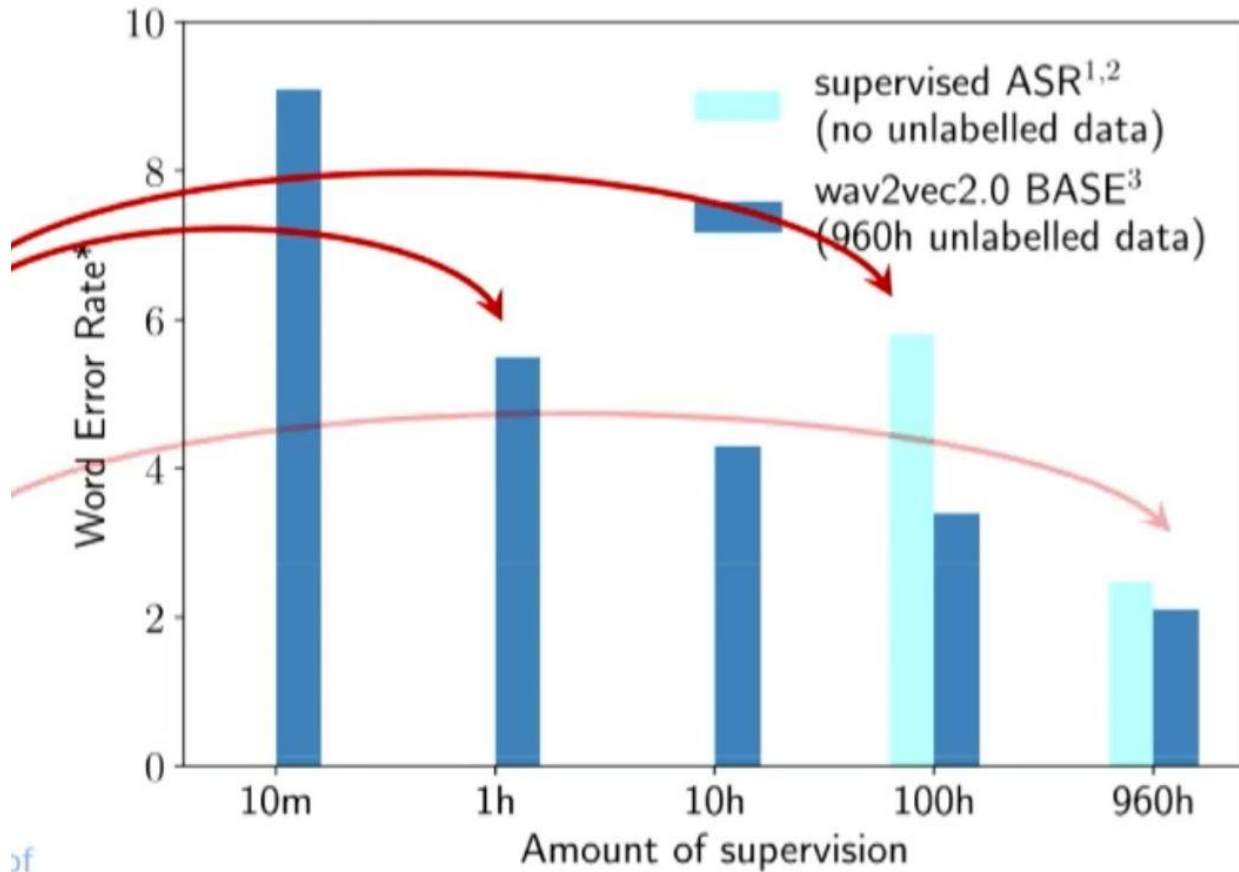
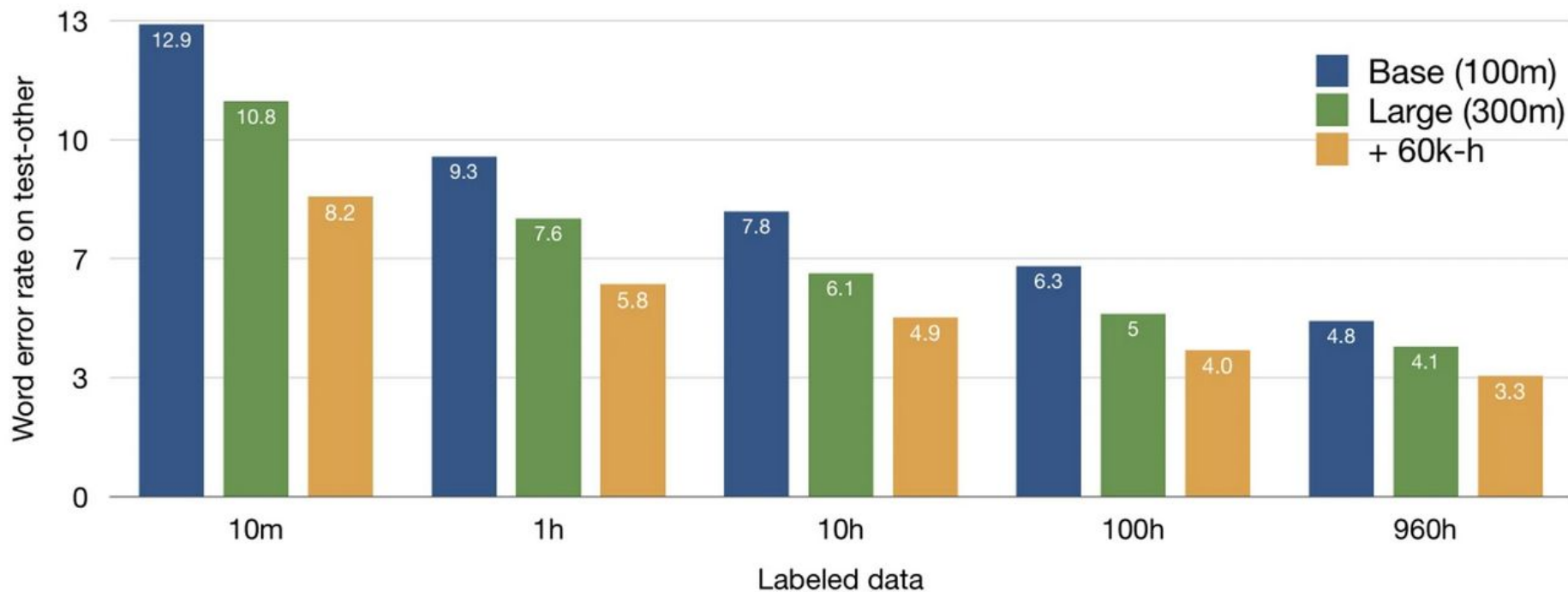


Figure 3: Visualization of the co-occurrence between discrete latent speech representations and phonemes. We plot the conditional probability $P(\text{phoneme}|\mathbf{q}_t)$ on TIMIT train data. The y-axis shows the collapsed 39 classes of phonemes and the x-axis is over the different discrete latents.



[1] Alexei Baevski, et. al, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations", arXiv, 2020.

Effects of model size and amount of unlabeled data



Wav2Vec2 is fine-tuned with transcribed data

Model	Unlabeled data	LM	dev		test	
			clean	other	clean	other
10 min labeled						
BASE	LS-960	None	46.1	51.5	46.9	50.9
		4-gram	8.9	15.7	9.1	15.6
		Transf.	6.6	13.2	6.9	12.9
LARGE	LS-960	None	43.0	46.3	43.5	45.3
		4-gram	8.6	12.9	8.9	13.1
		Transf.	6.6	10.6	6.8	10.8
LARGE	LV-60k	None	38.3	41.0	40.2	38.7
		4-gram	6.3	9.8	6.6	10.3
		Transf.	4.6	7.9	4.8	8.2
1h labeled						
BASE	LS-960	None	24.1	29.6	24.5	29.7
		4-gram	5.0	10.8	5.5	11.3
		Transf.	3.8	9.0	4.0	9.3
LARGE	LS-960	None	21.6	25.3	22.1	25.3
		4-gram	4.8	8.5	5.1	9.4
		Transf.	3.8	7.1	3.9	7.6
LARGE	LV-60k	None	17.3	20.6	17.2	20.3
		4-gram	3.6	6.5	3.8	7.1
		Transf.	2.9	5.4	2.9	5.8

10h labeled						
BASE	LS-960	None	10.9	17.4	11.1	17.6
		4-gram	3.8	9.1	4.3	9.5
		Transf.	2.9	7.4	3.2	7.8
LARGE	LS-960	None	8.1	12.0	8.0	12.1
		4-gram	3.4	6.9	3.8	7.3
		Transf.	2.9	5.7	3.2	6.1
LARGE	LV-60k	None	6.3	9.8	6.3	10.0
		4-gram	2.6	5.5	3.0	5.8
		Transf.	2.4	4.8	2.6	4.9

100h labeled						
BASE	LS-960	None	6.1	13.5	6.1	13.3
		4-gram	2.7	7.9	3.4	8.0
		Transf.	2.2	6.3	2.6	6.3
LARGE	LS-960	None	4.6	9.3	4.7	9.0
		4-gram	2.3	5.7	2.8	6.0
		Transf.	2.1	4.8	2.3	5.0
LARGE	LV-60k	None	3.3	6.5	3.1	6.3
		4-gram	1.8	4.5	2.3	4.6
		Transf.	1.9	4.0	2.0	4.0

Table 2: WER on Librispeech when using all 960 hours of labeled data (cf. Table 1).

Model	Unlabeled data	LM	dev		test	
			clean	other	clean	other
Supervised						
CTC Transf [51]	-	CLM+Transf.	2.20	4.94	2.47	5.45
S2S Transf. [51]	-	CLM+Transf.	2.10	4.79	2.33	5.17
Transf. Transducer [60]	-	Transf.	-	-	2.0	4.6
ContextNet [17]	-	LSTM	1.9	3.9	1.9	4.1
Conformer [15]	-	LSTM	2.1	4.3	1.9	3.9
Semi-supervised						
CTC Transf. + PL [51]	LV-60k	CLM+Transf.	2.10	4.79	2.33	4.54
S2S Transf. + PL [51]	LV-60k	CLM+Transf.	2.00	3.65	2.09	4.11
Iter. pseudo-labeling [58]	LV-60k	4-gram+Transf.	1.85	3.26	2.10	4.01
Noisy student [42]	LV-60k	LSTM	1.6	3.4	1.7	3.4
This work						
LARGE - from scratch	-	Transf.	1.7	4.3	2.1	4.6
BASE	LS-960	Transf.	1.8	4.7	2.1	4.8
LARGE	LS-960	Transf.	1.7	3.9	2.0	4.1
	LV-60k	Transf.	1.6	3.0	1.8	3.3

With self-training

Table 1: WER on the Librispeech dev and test sets for the Libri-light low-resource labeled data setups of 10 min, 1 hour and 10 hours. As unlabeled data we use the audio of Librispeech (LS-960) or the larger LibriVox (LV-60k). ST (s2s scratch) trains a sequence to sequence model with a word-piece vocabulary on the pseudo-labeled data from random initialization while as ST (ctc ft) fine-tunes wav2vec 2.0 with the pseudo-labels using CTC and a letter-based vocabulary. All results are with language models at inference time.

Model	Unlbl'd data	dev		test	
		clean	other	clean	other
10 min labeled					
Discr. BERT [27]	LS-960	15.7	24.1	16.3	25.2
wav2vec 2.0 [24]	LS-960	6.6	10.6	6.8	10.8
+ ST (s2s scratch)	LS-960	4.1	7.0	5.0	8.1
+ ST (ctc ft)	LS-960	3.6	6.6	4.0	7.2
wav2vec 2.0 [24]	LV-60k	5.0	8.4	5.2	8.6
+ ST (s2s scratch)	LV-60k	2.6	4.7	3.1	5.4
+ ST (ctc ft)	LV-60k	2.8	4.6	3.0	5.2
1h labeled					
Discr. BERT [27]	LS-960	8.5	16.4	9.0	17.6
wav2vec 2.0 [24]	LS-960	3.8	7.1	3.9	7.6
+ ST (s2s scratch)	LS-960	2.9	5.6	3.4	6.6
+ ST (ctc ft)	LS-960	2.8	5.5	3.1	6.3
10h labeled					
Discr. BERT [27]	LS-960	5.3	13.2	5.9	14.1
IPL [14]	LS-960	23.5	25.5	24.4	26.0
wav2vec 2.0 [24]	LS-960	2.9	5.7	3.2	6.1
+ ST (s2s scratch)	LS-960	2.5	5.1	3.5	5.9
+ ST (ctc ft)	LS-960	2.6	5.2	2.9	5.7

Wav2vec2 for Indian languages

- Finetuned on three different databases provided by IITM.
- Pretrained model: XSLR-53
 - Trained on 56000 hours of speech data of 53 different languages.

Language	Train (Hours)	Eval (Hours)	WER	LM (kenLM)
Indian English	179.5	5.4	4.91 %	6 Gram
Hindi	178.4	4.9	4.55 %	5 Gram
Tamil	104.5	3.8	5.84 %	4 Gram

- Kaldi based TDNN model

Language	Train (Hours)	Eval (Hours)	WER	LM (SRILM)
Indian English	179.5	5.4	4.97 %	RNNLM Rescore
Hindi	178.4	4.9	3.73 %	5 Gram
Tamil	104.5	3.8	5.21 %	5 Gram

Wav2vec2 pretrained model for Indian languages

CLSRIL-23: CROSS LINGUAL SPEECH REPRESENTATIONS FOR INDIC LANGUAGES

Anirudh Gupta[‡], Harveen Singh Chadha[‡], Priyanshi Shah¹, Neeraj Chhimwal¹, Ankur Dhuriya¹, Rishabh Gaur¹, and Vivek Raghavan²

¹Thoughtworks
²Ekstep Foundation

Paper link:
<https://arxiv.org/pdf/2107.07402.pdf>

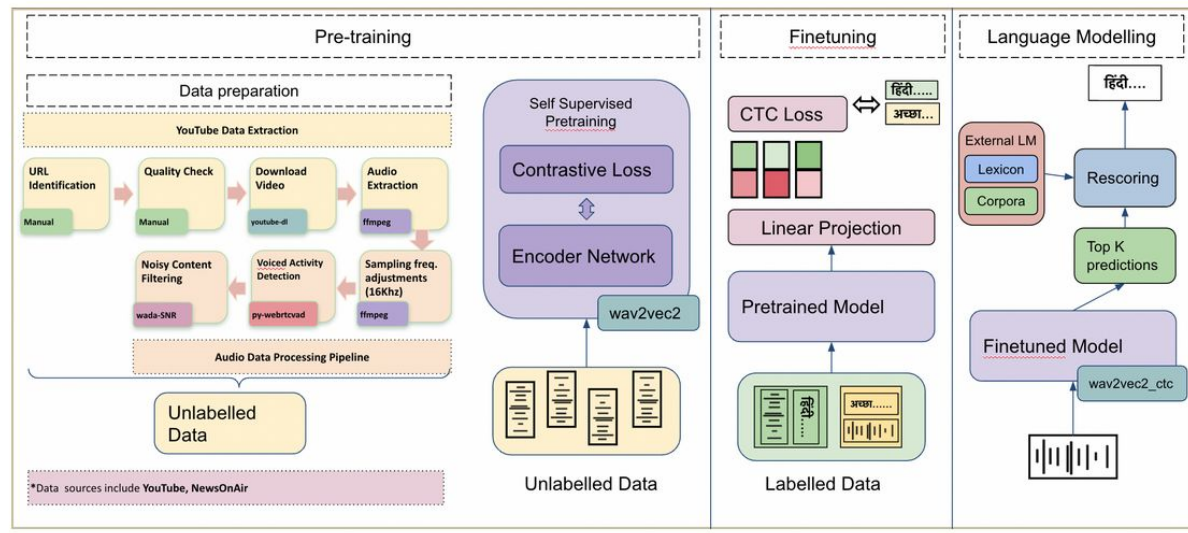
Model link:
<https://github.com/Open-Speech-EkStep/vakyansh-models>

Wav2vec2 pretrained model for Indian languages

IndicWav2Vec

IndicWav2Vec is a multilingual speech model pretrained on 40 Indian languages. This model represents the largest diversity of Indian languages in the pool of multilingual speech models. We fine-tune this model for downstream ASR for 9 languages and obtain state-of-the-art results on 3 public benchmarks, namely MUCS, MSR and OpenSLR.

As part of IndicWav2Vec we create largest publicly available corpora for 40 languages from 4 different language families. We also trained state-of-the-art ASR models for 9 Indian languages.



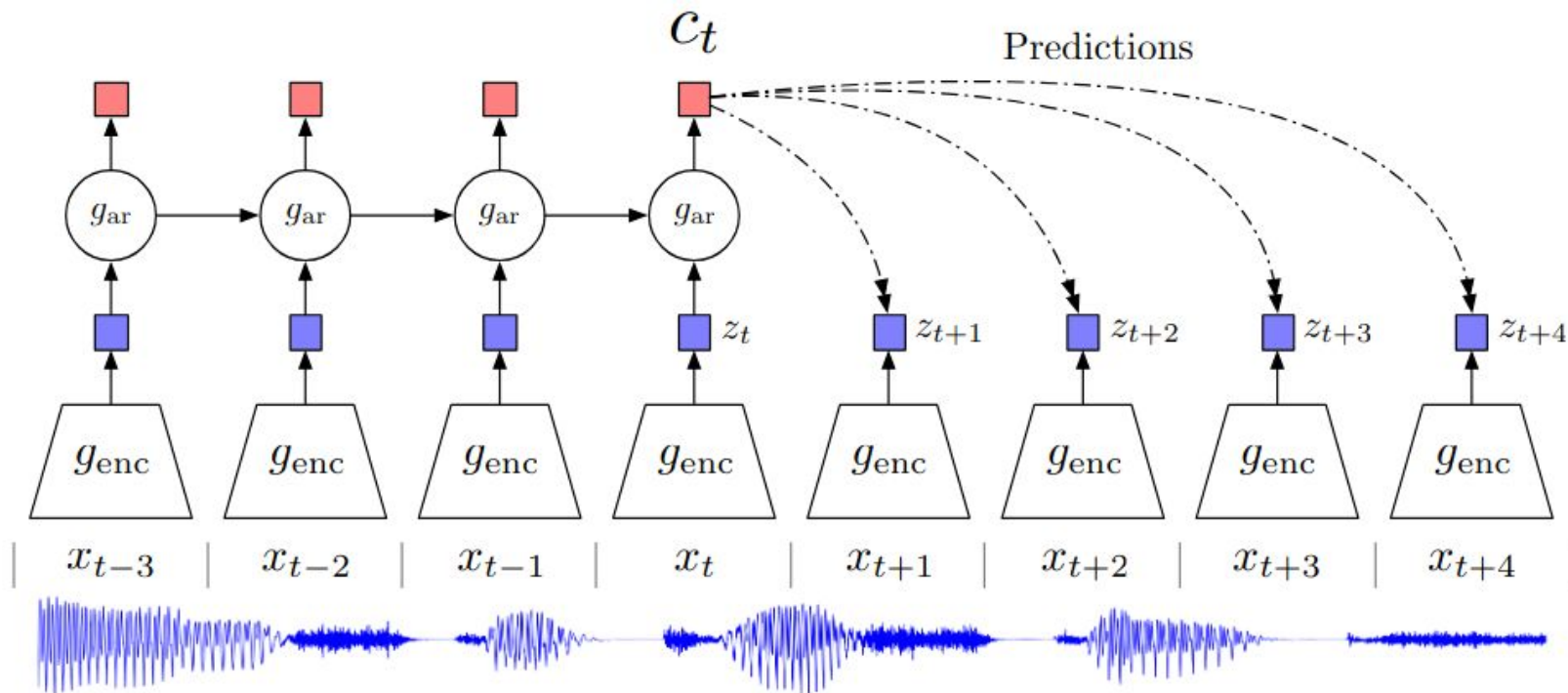
Paper link:

<https://ojs.aaai.org/index.php/AAAI/article/view/21327/21076>

Model link:

<https://github.com/AI4Bharat/IndicWav2Vec#download-models>

Contrastive Predictive Coding



LibriSpeech phone classification ablation experiments.

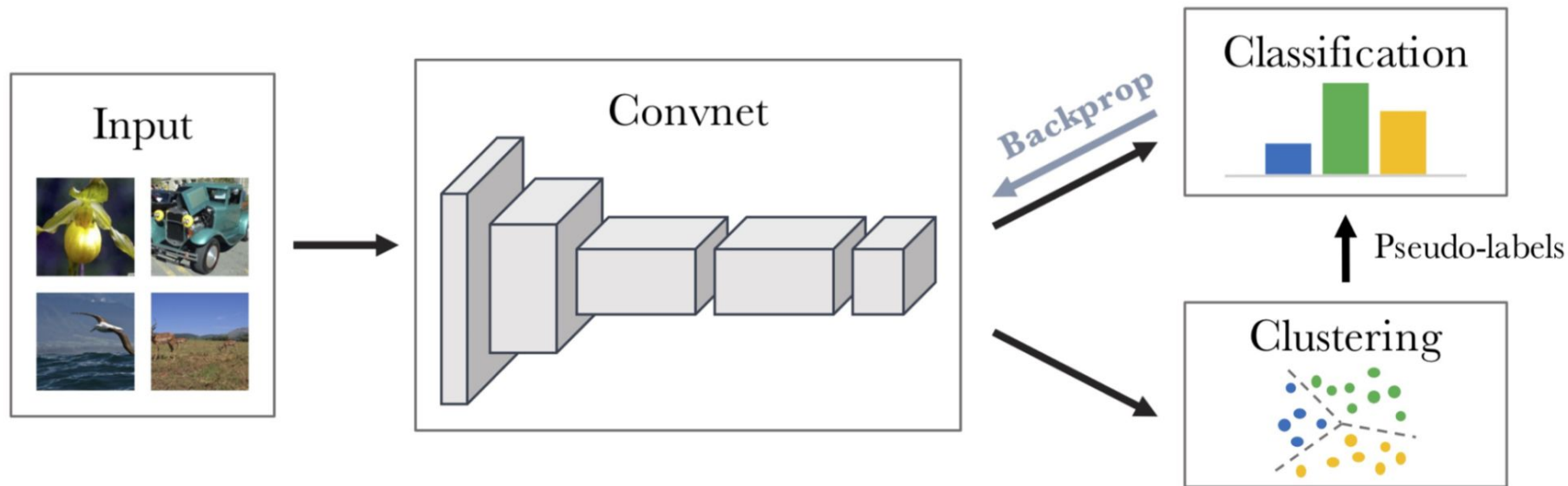
Comparing different strategies for drawing negative sample

Method	ACC
#steps predicted	
2 steps	28.5
4 steps	57.6
8 steps	63.6
12 steps	64.6
16 steps	63.8
Negative samples from	
Mixed speaker	64.6
Same speaker	65.5
Mixed speaker (excl.)	57.3
Same speaker (excl.)	64.6
Current sequence only	65.2

HuBERT (Hidden unit BERT)

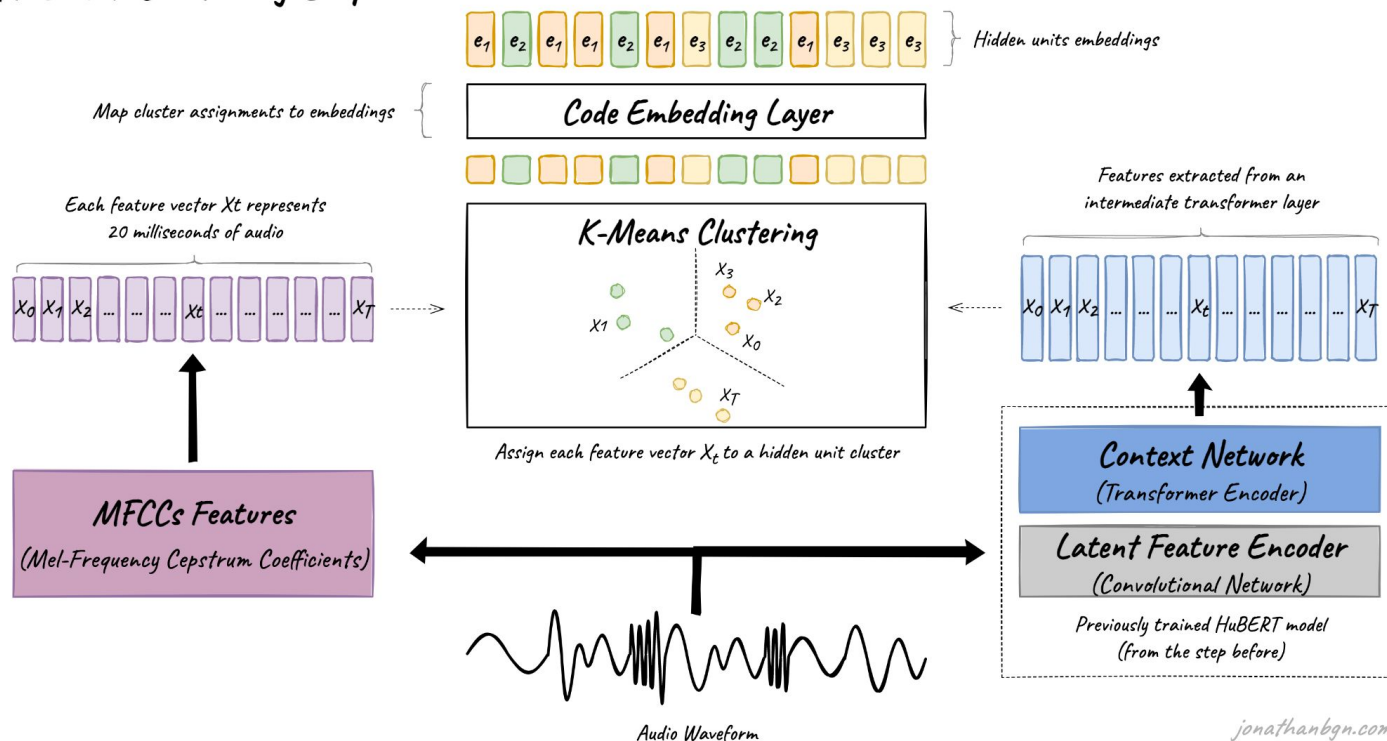
- Main idea to have an offline clustering step to provide aligned target labels for a BERT-like prediction loss
- HuBERT leverages a BERT-like architecture and uses an offline clustering step to generate noisy labels for masked language model pre-training, whereas wav2vec 2.0 learns its targets simultaneously during training using a quantization process.
- **HuBERT uses the cross-entropy loss**, instead of the more complex combination of contrastive loss + diversity loss used by wav2vec 2.0.

HuBERT draws inspiration from the [DeepCluster paper](#) in CV, where images are assigned to a given number of clusters before re-using these clusters as “pseudo-labels” for training the model in a self-supervised way.



Step 1: Discover “hidden units” targets through clustering

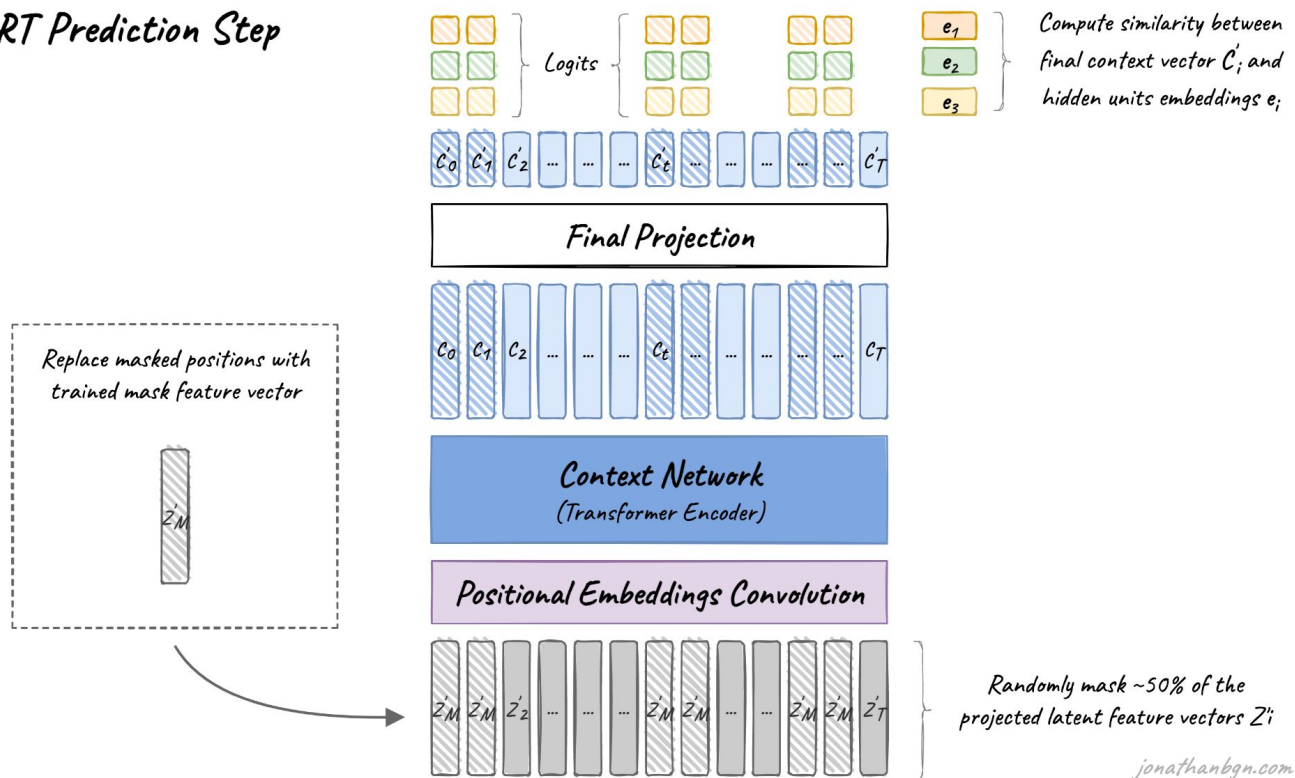
HuBERT Clustering Step



jonathanbgn.com

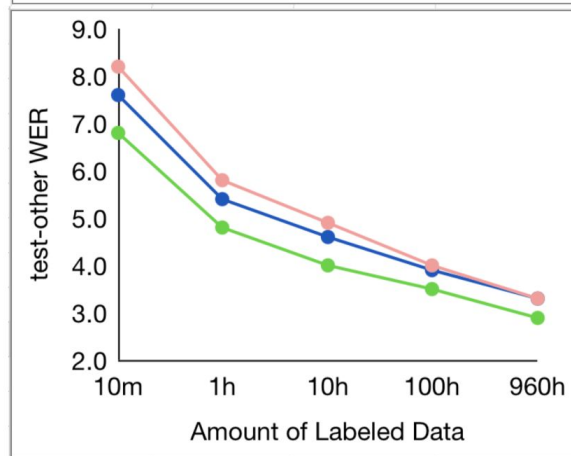
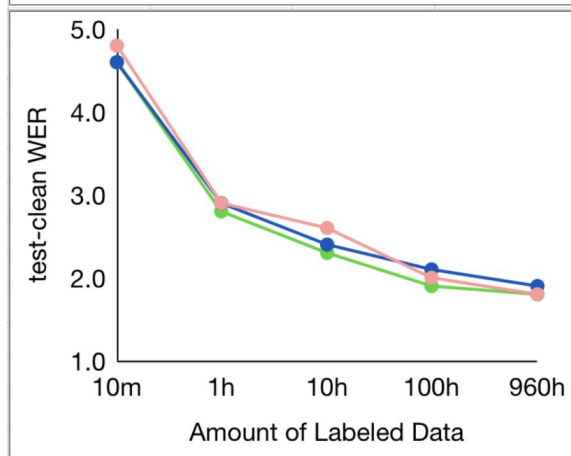
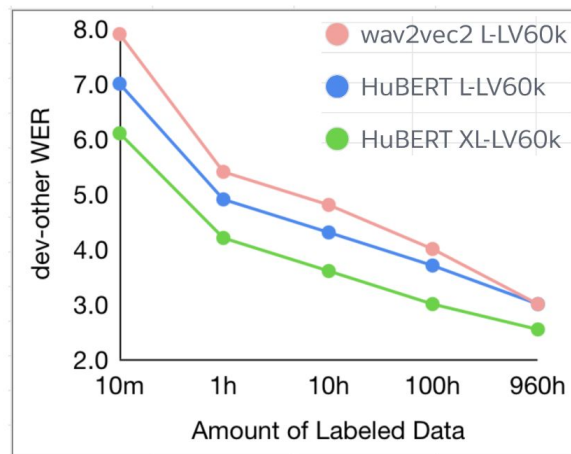
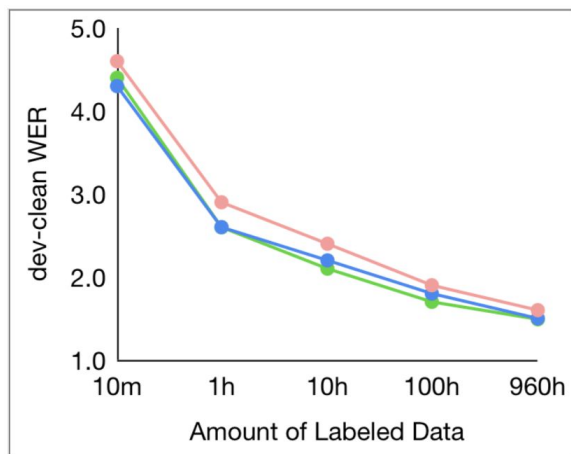
Step 2: Predict noisy targets from the context

HuBERT Prediction Step



		BASE	LARGE	X-LARGE
CNN Encoder	strides	5, 2, 2, 2, 2, 2, 2		
	kernel width	10, 3, 3, 3, 3, 2, 2		
	channel	512		
Transformer	layer	12	24	48
	embedding dim.	768	1024	1280
	inner FFN dim.	3072	4096	5120
	layerdrop prob	0.05	0	0
	attention heads	8	16	16
Projection	dim.	256	768	1024
Num. of Params		95M	317M	964M

TABLE I: Model architecture summary for BASE, LARGE, and X-LARGE HuBERT models



mHuBERT-147: A Compact Multilingual HuBERT Model

Marcely Zanon Boito¹, Vivek Iyer², Nikolaos Lagos¹, Laurent Besacier¹, Ioan Calapodescu¹

¹NAVER LABS Europe - France

²University of Edinburgh - UK

Training code: <https://github.com/utter-project/fairseq>

Pre-processing code: <https://github.com/utter-project/mHuBERT-147-scripts>

Model Weights: <https://huggingface.co/utter-project/mHuBERT-147>

SSL	# Params	Monolingual ASR	Multilingual ASR		LID	Multilingual ASR + LID			SUPERB _s (↑)
		normal CER (↓)	normal CER (↓)	few-shot CER (↓)	normal ACC (↑)	normal ACC (↑)	normal CER (↓)	few-shot CER (↓)	
MMS-1B	965M	33.3 / 25.7	21.3 / 18.1	30.2 / 30.8	84.8 / 86.1	73.3 / 74.8	26.0 / 25.5	25.4 / 24.8	983.5 🏆 / 948.1 🏆
mHuBERT-147	95M	34.2 / 26.3	23.6 / 22.0	33.2 / 32.9	85.3 / 91.0	81.4 / 90.0	26.2 / 22.1	34.9 / 33.5	949.8 🏆 / 950.2 🏆
MMS-300M	317M	33.8 / 30.5	28.7 / 24.0	36.5 / 36.5	62.3 / 84.3	71.9 / 74.3	31.5 / 30.0	30.9 / 29.2	824.9 🏆 / 844.3
NWHC1	317M	39.5 / 30.5	28.9 / 21.5	41.4 / 38.6	67.1 / 87.4	77.1 / 90.6	28.8 / 21.5	40.3 / 38.2	774.4 / 876.9 🏆
NWHC2	317M	39.5 / 30.5	29.3 / 21.6	42.0 / 39.3	64.4 / 88.1	77.4 / 90.6	28.4 / 21.8	41.5 / 38.8	759.9 / 873.3
XLS-R-300M	317M	39.7 / 30.6	29.2 / 22.0	40.9 / 39.3	66.9 / 87.9	55.6 / 85.6	28.4 / 22.9	42.1 / 42.4	730.8 / 850.5
WavLabLM-large-MS	317M	40.5 / 32.8	37.8 / 31.9	43.8 / 42.8	71.7 / 81.1	70.8 / 80.0	37.0 / 32.2	43.4 / 41.2	707.5 / 740.9

Table 1: ML-SUPERB 10min/1h leaderboard. Updated SOTA scores for each metric are presented in **bold**. The first (🏆), second (🥈) and third (🥉) best SUPERB scores are highlighted.

WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing

Sanyuan Chen*, Chengyi Wang*, Zhengyang Chen*, Yu Wu*, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, Furu Wei

WavLM is built based on the HuBERT framework, with an emphasis on both spoken content modeling and speaker identity preservation.

WavLM jointly learns masked speech prediction and denoising in pre-training.

For better speaker discrimination, an utterance mixing training strategy, where additional overlapped utterances are created unsupervisely and incorporated during model training.

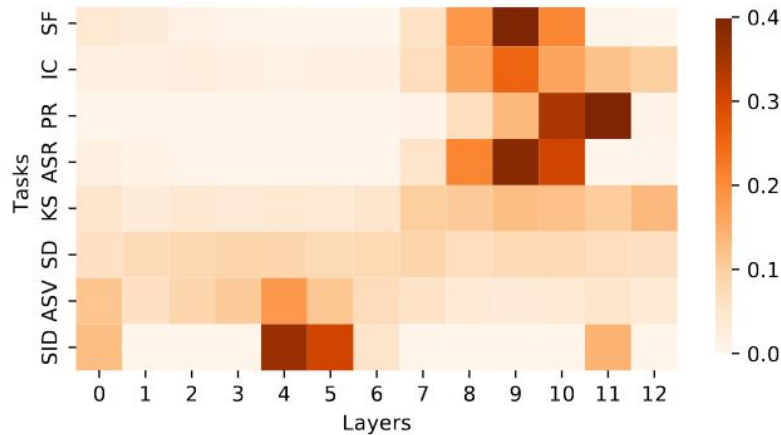
Utterance mix strategy

We simulate the noisy speech with multiple speakers and various background noise for self-supervised pre-training.

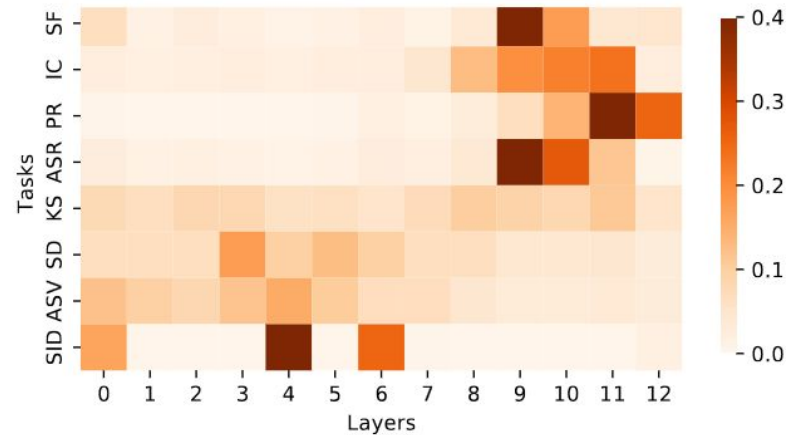
During training, the model still does a masked prediction task (predict masked frames), but now from **mixed** audio instead of single-speaker data.

By hearing multiple speakers or noise overlaid, the model learns to tease apart (or become robust to) multi-talker scenarios, background noise, and other real-world complexities.

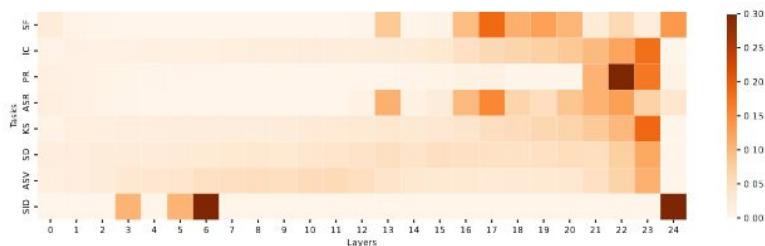
This leads to better performance on tasks like speaker diarization, speech recognition in noisy conditions, etc.



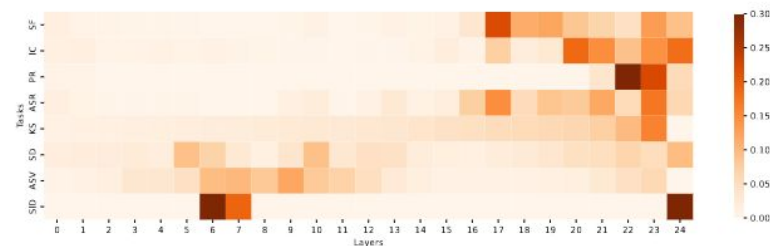
(a) HuBERT Base



(b) WavLM Base+



(c) HuBERT Large



(d) WavLM Large

Fig. 2. Weight analysis on the SUPERB Benchmark. Layer 0 corresponds to the input of the first Transformer layer. The y-axis represents different tasks, while the x-axis represents different layers.

TABLE I

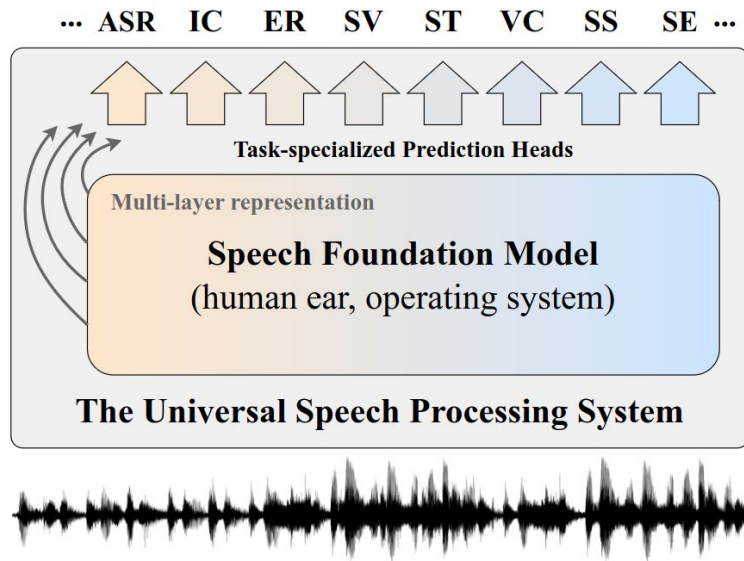
UNIVERSAL SPEECH REPRESENTATION EVALUATION ON SUPERB BENCHMARK. PARAL DENOTE PARALINGUISTICS ASPECT OF SPEECH.

Method	#Params	Corpus	Speaker			Content					Semantics				ParaL	Generation					Overall	
			SID	ASV	SD	PR	ASR	OOD-ASR	KS	QoE	ST	IC	SF	ER	SE	SS	VC					
			Acc ↑	EER ↓	DER ↓	PER ↓	WER ↓	WER ↓	Acc ↑	MTWV ↑	BLEU ↑	Acc ↑	F1 ↑	CER ↓	Acc ↑	PESQ ↑	STOI ↑	SI-SDR ↑	MCD ↓	WER ↓		ASV ↑
FBANK	0	-	8.5E-4	9.56	10.05	82.01	23.18	63.58	8.63	0.0058	2.32	9.10	69.64	52.94	35.39	2.55	93.6	9.23	8.47	38.3	77.25	43.2
PASE+ [44]	7.83M	LS 50 hr	37.99	11.61	8.68	58.87	25.11	61.56	82.54	0.0072	3.16	29.82	62.14	60.17	57.86	2.56	93.9	9.87	8.66	30.6	63.20	51.5
APC [30]	4.11M	LS 360 hr	60.42	8.56	10.53	41.98	21.28	63.12	91.01	0.0310	5.95	74.69	70.46	50.89	59.33	2.56	93.4	8.92	8.05	27.2	87.25	59.2
VQ-APC [29]	4.63M	LS 360 hr	60.15	8.72	10.45	41.08	21.20	63.56	91.11	0.0251	4.23	74.48	68.53	52.91	59.66	2.56	93.4	8.44	7.84	22.4	94.25	59.5
NPC [33]	19.38M	LS 360 hr	55.92	9.40	9.34	43.81	20.20	61.66	88.96	0.0246	4.32	69.44	72.79	48.44	59.08	2.52	93.1	8.04	7.86	30.4	94.75	59.0
Mockingjay [35]	85.12M	LS 360 hr	32.29	11.66	10.54	70.19	22.82	65.27	83.67	6.6E-04	4.45	34.33	61.59	58.89	50.28	2.53	93.4	9.29	8.29	35.1	79.75	51.0
TERA [34]	21.33M	LS 960 hr	57.57	15.89	9.96	49.17	18.17	58.49	89.48	0.0013	5.66	58.42	67.50	54.17	56.27	2.54	93.6	10.19	8.21	25.1	83.75	57.2
DeCoAR 2.0 [37]	89.84M	LS 960 hr	74.42	7.16	6.59	14.93	13.02	53.62	94.48	0.0406	9.94	90.80	83.28	34.73	62.47	2.47	93.2	8.54	7.83	17.1	90.75	66.3
modified CPC [53]	1.84M	LL 60k hr	39.63	12.86	10.38	42.54	20.18	62.54	91.88	0.0326	4.82	64.09	71.19	49.91	60.96	2.57	93.7	10.40	8.41	26.2	71.00	56.9
wav2vec [39]	32.54M	LS 960 hr	56.56	7.99	9.9	31.58	15.86	55.86	95.59	0.0485	6.61	84.92	76.37	43.71	59.79	2.53	93.8	9.30	7.45	10.1	98.25	63.5
vq-wav2vec [40]	34.15M	LS 960 hr	38.80	10.38	9.93	33.48	17.71	60.66	93.38	0.0410	5.66	85.68	77.68	41.54	58.24	2.48	93.6	8.16	7.08	13.4	100.00	61.8
wav2vec 2.0 Base [5]	95.04M	LS 960 hr	75.18	6.02	6.08	5.74	6.43	46.95	96.23	0.0233	14.81	92.35	88.30	24.77	63.43	2.55	93.9	9.77	7.50	10.5	98.00	69.6
HuBERT Base [6]	94.68M	LS 960 hr	81.42	5.11	5.88	5.41	6.42	46.69	96.30	0.0736	15.53	98.34	88.53	25.20	64.92	2.58	93.9	9.36	7.47	8.0	98.50	70.9
WavLM Base	94.70M	LS 960 hr	84.51	4.69	4.55	4.84	6.21	42.81	96.79	0.0870	20.74	98.63	89.38	22.86	65.94	2.58	94.0	10.37	7.42	8	98.00	72.0
- w/o denoising task	94.70M	LS 960 hr	84.39	4.91	6.03	4.85	6.08	43.61	96.79	0.0799	21.03	98.42	88.69	23.43	65.55	2.56	93.9	9.91	7.43	7.5	97.75	71.7
- w/o structure modification	94.68M	LS 960 hr	84.74	4.61	4.72	5.22	6.80	42.88	96.79	0.0956	20.03	98.31	88.56	24.00	65.60	2.58	94.0	10.29	7.45	8.4	99.00	71.9
WavLM Base+	94.70M	Mix 94k hr	89.42	4.07	3.50	3.92	5.59	38.32	97.37	0.0988	24.25	99.00	90.58	21.20	68.65	2.63	94.3	10.85	7.40	8.1	99.00	73.4
wav2vec 2.0 Large [5]	317.38M	LL 60k hr	86.14	5.65	5.62	4.75	3.75	44.69	96.66	0.0489	12.48	95.28	87.11	27.31	65.64	2.52	94.0	10.02	7.63	15.8	97.25	70.4
HuBERT Large [6]	316.61M	LL 60k hr	90.33	5.98	5.75	3.53	3.62	44.08	95.29	0.0353	20.01	98.76	89.81	21.76	67.62	2.64	94.2	10.45	7.22	9.0	99.25	72.2
WavLM Large	316.62M	Mix 94k hr	95.49	3.77	3.24	3.06	3.44	32.27	97.86	0.0886	26.57	99.31	92.21	18.36	70.62	2.70	94.5	11.19	7.30	9.0	99.00	74.6

A Large-Scale Evaluation of Speech Foundation Models

Shu-wen Yang, Heng-Jui Chang*, Zili Huang*, Andy T. Liu*, Cheng-I Lai*, Haibin Wu*, Jiatong Shi, Xuankai Chang, Hsiang-Sheng Tsai, Wen-Chin Huang, Tzu-hsun Feng, Po-Han Chi, Yist Y. Lin, Yung-Sung Chuang, Tzu-Hsien Huang, Wei-Cheng Tseng, Kushal Lakhotia, Shang-Wen Li, Abdelrahman Mohamed, Shinji Watanabe, Hung-yi Lee

Abstract—The foundation model paradigm leverages a shared foundation model to achieve state-of-the-art (SOTA) performance for various tasks, requiring minimal downstream-specific modeling and data annotation. This approach has proven crucial in the field of Natural Language Processing (NLP). However, the speech processing community lacks a similar setup to explore the paradigm systematically. In this work, we establish the Speech processing Universal PERFORMANCE Benchmark (SUPERB) to study the effectiveness of the paradigm for speech. We propose a unified multi-tasking framework to address speech processing tasks in SUPERB using a frozen foundation model followed by task-specialized, lightweight prediction heads. Combining our results with community submissions, we verify that the foundation model paradigm is promising for speech, and our multi-tasking framework is simple yet effective, as the best-performing foundation model shows competitive generalizability across most SUPERB tasks. For reproducibility and extensibility, we have developed a long-term maintained platform that enables deterministic benchmarking, allows for result sharing via



Speech processing Universal PERformance Benchmark (SUPERB)

- SUPERB standardizes speech SSL evaluation with a broad range of 15 speech processing tasks.
- Phoneme recognition (PR), keyword spotting (KS), speaker identification (SID), emotion recognition (ER), intent classification (IC), slot filling (SF), automatic speech recognition (ASR), speaker verification (SV), speaker diarization (SD), query-by-example spoken term detection (QbE), speech translation (ST), out-of-domain ASR (OOD-ASR), source separation (SS), speech enhancement (SE), and voice conversion (VC).

<https://github.com/s3prl/s3prl/blob/main/s3prl/downstream/docs/superb.md>

Model	#Params	Corpus	Implementation (GitHub)	Use official checkpoint	Community submission	Tasks underperforming FBANK
FBANK	-	-	pytorch/audio [101]	-	-	-
PASE+ [16]	7.83M	LS 50 hr	santi-pdp/pase	✓	×	ASR, SF, SV, VC
APC [8]	4.11M	LS 360 hr	iamyuanchung/Autoregressive-Predictive-Coding	×	×	SD, SS
VQ-APC [102]	4.63M	LS 360 hr	Alexander-H-Liu/NPC	×	×	SF, SD, SS
NPC [103]	19.38M	LS 360 hr	Alexander-H-Liu/NPC	×	×	SE, SS
Mockingjay [9]	85.12M	LS 360 hr	s3prl/s3prl	✓	×	OOD-ASR, SF, QbE, SV, SD, SE
TERA [10]	21.33M	LS 960 hr		✓	×	
Audio Albert [104]	7.15M	LS 960 hr		✓	×	QbE
DistilHuBERT [105]	27.03M	LS 960 hr		✓	✓	SS
DeCoAR [106]	67.25M	LS 960 hr	awslabs/speech-representations	✓	×	-
DeCoAR 2.0 [11]	89.84M	LS 960 hr		✓	×	SE, SS
Modified CPC [107]	1.84M	LL 60k hr	facebookresearch/CPC_audio	✓	×	SV, SD
data2vec-aqc Base [108]	93.84M	LS 960 hr	Speech-Lab-IITM/data2vec-aqc	✓	✓	-
CCC-wav2vec 2.0 Base [109]	95.04M	LS 960 hr	Speech-Lab-IITM/CCC-wav2vec-2.0	✓	✓	-
FaST-VGS+ [110]	217.23M	LS 960 hr SC 742 hr	jasonppy/FaST-VGS-Family	✓	✓	-
LightHuBERT Stage1 [111]	94.38M	LS 960 hr	mechanicalsea/lighthubert	✓	✓	-
LightHuBERT Small [111]	26.88M	LS 960 hr		✓	✓	SE
CoBERT Base [112]	94.35M	LS 960 hr	mct10/CoBERT	✓	✓	SE, SS
wav2vec [13]	32.54M	LS 960 hr	pytorch/fairseq [51]	✓	×	SE
vq-wav2vec [14]	32.15M	LS 960 hr		✓	×	SV, SE, SS
wav2vec 2.0 Base [15]	95.04M	LS 960 hr		✓	×	-
wav2vec 2.0 Large [15]	317.38M	LL 60k hr		✓	×	SE
HuBERT Base [17]	94.68M	LS 960 hr		✓	×	-
HuBERT Large [17]	316.61M	LL 60k hr		✓	×	-
Data2vec Base [25]	93.16M	LS 960 hr		✓	✓	SE, SS
Data2vec Large [25]	313.28M	LL 60k hr		✓	✓	-
DPHuBERT [113]	23.59M	LS 960hr	pyf98/DPHuBERT	✓	✓	-
DPWavLM [113]	23.59M	LS 960hr		✓	✓	SS
Unispeech SAT Base [114]	94.37M	LS 960 hr	microsoft/UniSpeech	✓	×	-
Unispeech SAT Base+ [114]	94.37M	LVG 94k hr		✓	×	-
Unispeech SAT Large [114]	315.43M	LVG 94k hr		✓	×	-
WavLM Base [18]	94.38M	LS 960 hr		✓	✓	-
WavLM Base+ [18]	94.38M	LVG 94k hr		✓	✓	-
WavLM Large [18]	315.45M	LVG 94k hr		✓	✓	-

- fbank as the baseline**

Phonetic

Speaker

Semantic

Emotion

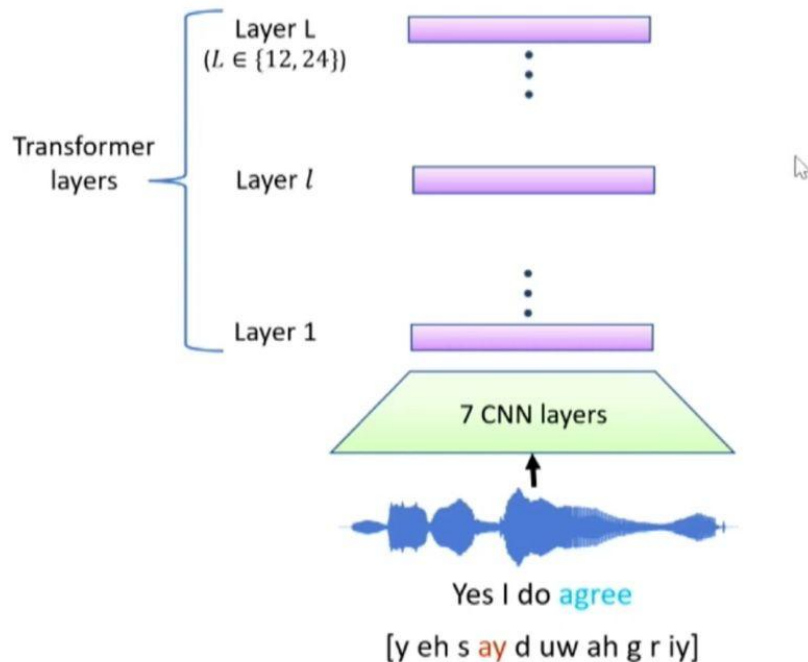
	PR	KS	ASR	QbE	SID	ASV	SD	IC	SF	ER
fbank	82.01	8.63	15.21	0.0058	8.50E-04	9.56	10.05	9.1	69.64	35.39
PASE+	58.87	82.54	16.62	0.0072	37.99	11.61	8.68	29.82	62.14	57.86
APC	41.98	91.01	14.74	0.0310	60.42	8.56	10.53	74.69	70.46	59.33
VQ-APC	41.08	91.11	15.21	0.0251	60.15	8.72	10.45	74.48	68.53	59.66
NPC	43.81	88.96	13.91	0.0246	55.92	9.40	9.34	69.44	72.79	59.08
Mockingjay	70.19	83.67	15.48	6.60E-04	32.29	11.66	10.54	34.33	61.59	50.28
TERA	49.17	89.48	12.16	0.0013	57.57	15.89	9.96	58.42	67.50	56.27
DeCoAR 2.0	14.93	94.48	9.07	0.0406	74.42	7.16	6.59	90.80	83.28	62.47
modified CPC	42.54	91.88	13.53	0.0326	39.63	12.86	10.38	64.09	71.19	60.96
wav2vec	31.58	95.59	11.00	0.0485	56.56	7.99	9.90	84.92	76.37	59.79
vq-wav2vec	33.48	93.38	12.80	0.0410	38.80	10.38	9.93	85.68	77.68	58.24
wav2vec 2.0 base	5.74	96.23	4.79	0.0233	75.18	6.02	6.08	92.35	88.30	63.43
wav2vec 2.0 large	4.75	96.66	3.10	0.0489	86.14	5.65	5.62	95.28	87.11	65.64
HuBERT base	5.41	96.30	4.79	0.0736	81.42	5.11	5.88	98.34	88.53	64.92
HuBERT large	3.53	95.29	2.94	0.0353	90.33	5.98	5.75	98.76	89.81	67.62

TABLE IV: The last layer representation v.s. weighted-sum over all layers. In each cell, the upper number represents the last layer; the lower number represents the weighted-sum. Bold fonts highlight the cases when weighted-sum is worse.

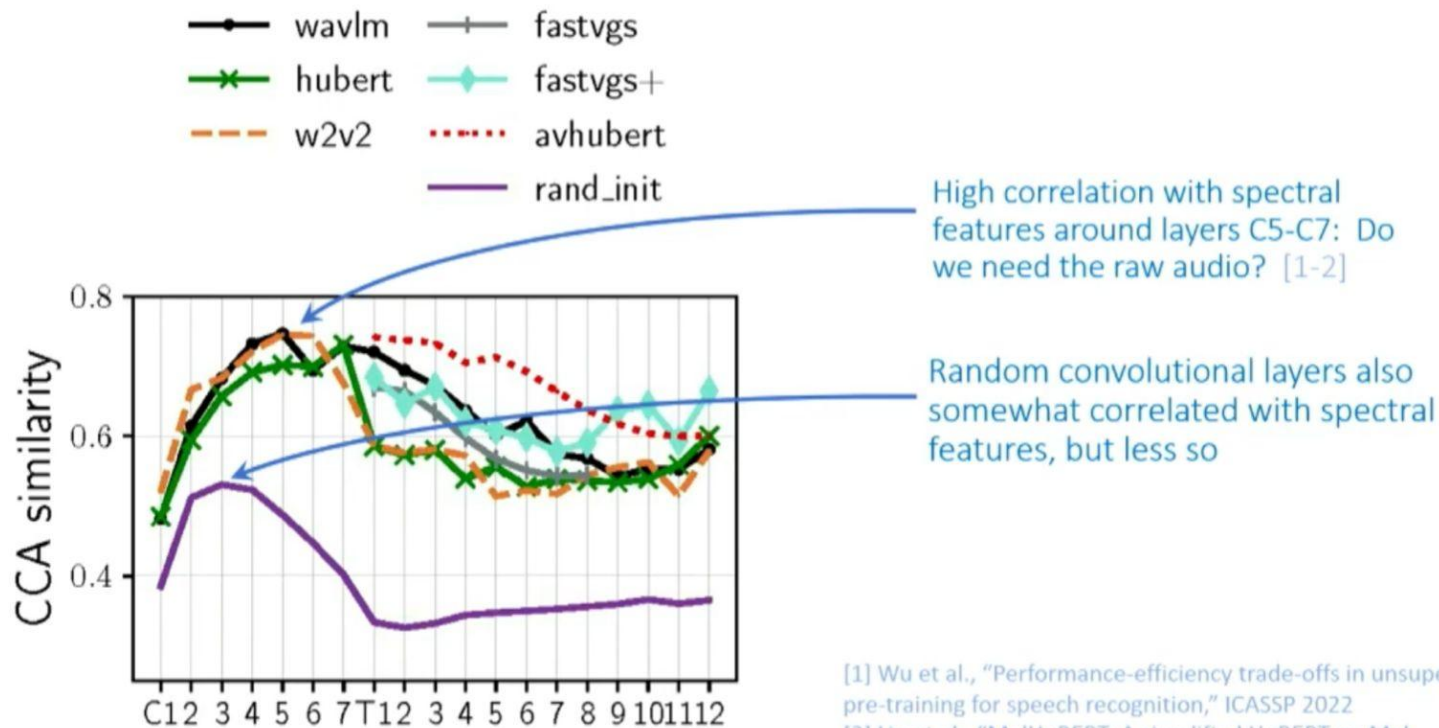
Models	PR	KS	SID	IC	ER
	per ↓	acc ↑	acc ↑	acc ↑	acc ↑
PASE+ [16]	58.88 58.87	82.37 82.54	35.84 37.99	30.29 29.82	57.64 57.86
APC [8]	41.85 41.98	91.04 91.01	59.79 60.42	74.64 74.69	58.84 59.33
VQ-APC [102]	42.86 41.08	90.52 91.11	49.57 60.15	70.52 74.48	58.31 59.66
TERA [10]	47.53 49.17	88.09 89.48	58.67 57.57	48.80 58.42	54.76 56.27
wav2vec [13]	32.39 31.58	94.09 95.59	44.88 56.56	78.91 84.92	58.17 59.79
vq-wav2vec [14]	53.49 33.48	92.28 93.38	39.04 38.80	59.4 85.68	55.89 58.24
wav2vec 2.0 Base [15]	28.37 5.74	92.31 96.23	45.62 75.18	58.34 92.35	56.93 63.43
HuBERT Base [17]	6.85 5.41	95.98 96.30	64.84 81.42	95.94 98.34	62.94 64.92

Layer-wise analysis:

Method: Extract layer-wise representations, and measure their similarity to external linguistic variables



Similarity with Mel spectral features



- [1] Wu et al., "Performance-efficiency trade-offs in unsupervised pre-training for speech recognition," ICASSP 2022
- [2] Lin et al., "MelHuBERT: A simplified HuBERT on Mel spectrogram," arXiv:2211.09944

Layer-wise analysis of a self-supervised speech representation

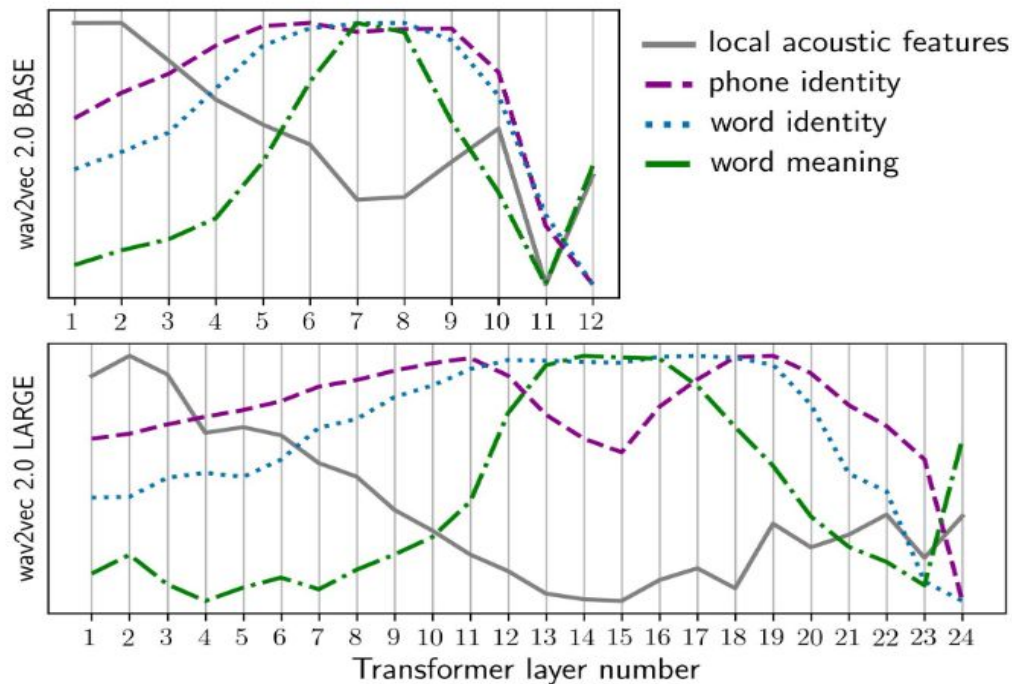


Image source: https://homepages.inf.ed.ac.uk/htang2/sigml/mlslp2021/MLSLP2021_paper_15.pdf

Layer-wise analysis of depression detection accuracy (%) of HuBERT and WavLM

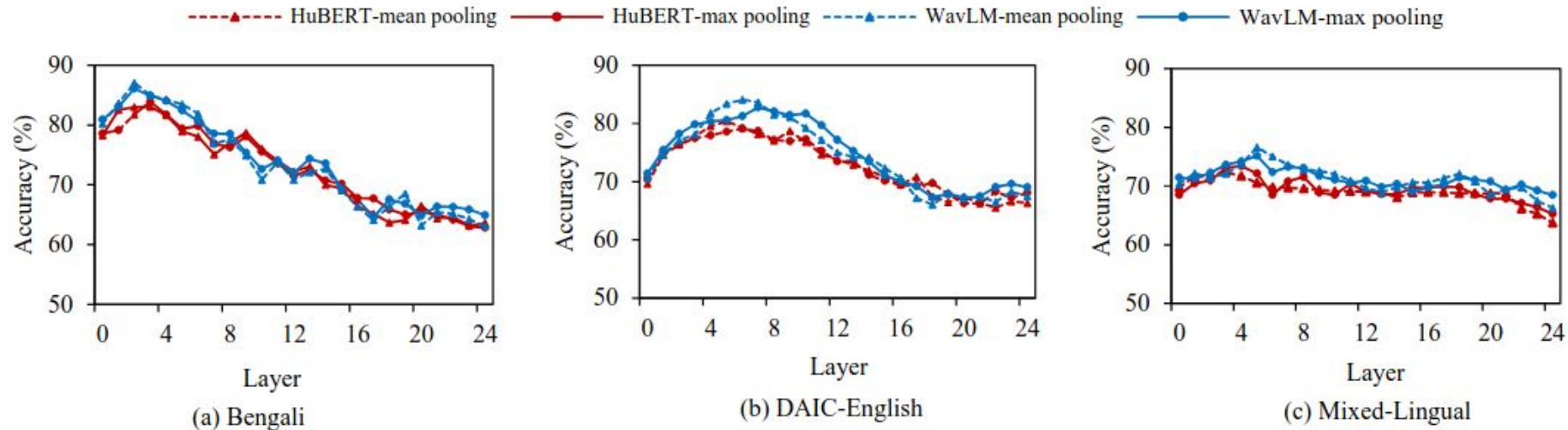


Image source: https://www.isca-archive.org/interspeech_2024/maji24_interspeech.pdf

Analyzing Domain Shift in SSL-based Pre-Training - 1

Domain shift: data distribution mismatch between training and testing

Domain mismatch between
the **unlabeled data for**
pre-training, **the labeled**
domain

Much easier to obtain
unlabeled target domain
data than labeled data

Findings: Adding unlabeled data
whose domain matches the test data
always improves performance, even if
the labeled data for fine-tuning is
out-of-domain.

Analyzing Domain Shift in SSL-based Pre-Training - 2

HOW DOES PRE-TRAINED WAV2VEC 2.0 PERFORM ON DOMAIN-SHIFTED ASR? AN EXTENSIVE BENCHMARK ON AIR TRAFFIC CONTROL COMMUNICATIONS

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Model (num. params.)	Unlabeled data	NATS		ISAVIA		ATCO2-Test		LiveATC-Test		LS*
		Greedy	+LM	Greedy	+LM	Greedy	+LM	Greedy	+LM	-
Baseline (31M)										
Hybrid-based [†]	-	-	7.7	-	12.5	-	24.7	-	35.8	-
BASE (95M)										
w2v2-B	LS	10.7	8.4	12.5	10.1	45.6	40.1	48.1	42.2	7.8
LARGE (371M)										
w2v2-L	LS	9.3	7.6	11.7	9.5	44.9	40.0	47.5	41.4	6.1
w2v2-L-60k	LS+LV	6.8	5.4	8.8	7.3	34.6	31.2	39.8	34.5	4.9
w2v2-L-60k+ ^{††}	LS+LV	9.3	<u>7.4</u>	11.2	9.1	23.3	21.2	31.1	27.2	-
XLS-R (300M)										
w2v2-XLS-R	ML	8.4	6.5	10.5	8.2	39.1	33.8	42.9	36.7	15.4
w2v2-XLS-R+ ^{††}	ML	<u>9.0</u>	<u>7.4</u>	<u>10.4</u>	<u>8.3</u>	<u>22.8</u>	<u>19.8</u>	<u>29.7</u>	<u>24.9</u>	-